Math 214: Introduction to Topology

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Conventions

- 2^X denotes the power set of X; $\mathbb{N} = \{1, 2, 3, \dots\}$.
- A *neighbourhood* of a point x means an *open* set containing x, unless stated otherwise. In Section 14.6 a "compact neighbourhood" of x is a compact set C with $x \in \operatorname{int}(C)$.
- $\operatorname{cl}(E)$ and $\overline{E}$ both denote closure; $\operatorname{int}(E)$ and E° both denote interior. When the ambient space matters we write $\operatorname{cl}_Y(E)$.
- In a metric space (X, d), $B_\varepsilon(x) = \{y \in X : d(x, y) < \varepsilon\}$.
- $X \cong Y$ means X and Y are homeomorphic.

Chapter 1

Motivation

1.1 Fixed point theorems

Theorem 1.1. *Let $f\colon [0,1] \to [0,1]$ be continuous. Then there is $x \in [0,1]$ with $f(x) = x$.*

Proof. Let $g(x) = f(x) - x$, which is continuous. Since $f(0) \geq 0$ we have $g(0) \geq 0$, and since $f(1) \leq 1$ we have $g(1) \leq 0$. By the intermediate value theorem there is $x_0 \in [0,1]$ with $g(x_0) = 0$, i.e. $f(x_0) = x_0$. □

Theorem 1.2 (Brouwer's fixed point theorem)**.** *Every continuous map $f\colon D \to D$ of the closed disk $D = \{x \in \mathbb{R}^2 : \|x\| \leq 1\}$ has a fixed point.*

Remark 1.3. Theorem 1.1 is the one-dimensional case. The proof of Brouwer's theorem in dimension 2 needs genuinely topological tools (e.g. the fundamental group), which is one motivation for the course: the statement only involves *continuity*, so the proof should only use topological information.

1.2 Euler's formula

Theorem 1.4 (Euler's formula)**.** *Let G be a connected graph drawn in the plane without edge crossings, with V vertices, E edges and F faces (including the unbounded face). Then*

$$V - E + F = 2.$$

1.3 Euclidean spaces

Let $X = \mathbb{R}^d$ with the Euclidean distance d.

- A sequence $x_1, x_2, \ldots$ *converges* to x if

$$\forall \varepsilon > 0\ \exists n_0\ \forall n \geq n_0 :\ d(x_n, x) < \varepsilon, \qquad \text{i.e.} \qquad x_n \in B_\varepsilon(x) \text{ eventually.}$$

- A map $f\colon \mathbb{R}^d \to \mathbb{R}^m$ is *continuous at* x if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x' :\ d(x, x') < \delta \implies d(f(x), f(x')) < \varepsilon,$$

  i.e. $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$. Equivalently, $x_n \to x$ implies $f(x_n) \to f(x)$.

Both definitions only refer to balls. The idea of topology is to keep only the information "which sets are open" and forget the distance.

Chapter 2

Topological Spaces

Reading: Munkres, *Topology* (2nd ed.), §§12, 17, 18.

2.1 Open sets in $\mathbb{R}^d$

Definition 2.1. Let $X = \mathbb{R}^d$.

- $U \subseteq X$ is *open* if $\forall x \in U\ \exists \varepsilon > 0 :\ B_\varepsilon(x) \subseteq U$.
- $V \subseteq X$ is *closed* if $X \setminus V$ is open.
- The *closure* of A is $\mathrm{cl}(A) = \{x \in \mathbb{R}^d : \forall \varepsilon > 0,\ B_\varepsilon(x) \cap A \neq \varnothing\}$.
- The *interior* of A is $\mathrm{int}(A) = \{x \in \mathbb{R}^d : \exists \varepsilon > 0,\ B_\varepsilon(x) \subseteq A\}$.

2.2 Axioms of topology

Definition 2.2 (Topology)**.** A *topology* on a set X is a family $\mathcal{T} \subseteq 2^X$ (whose members are called *open sets*) such that

(1) $\varnothing, X \in \mathcal{T}$;

(2) an arbitrary union of open sets is open: $\{U_i\}_{i \in I} \subseteq \mathcal{T} \implies \bigcup_{i \in I} U_i \in \mathcal{T}$;

(3) a finite intersection of open sets is open: $U_1, \ldots, U_n \in \mathcal{T} \implies U_1 \cap \cdots \cap U_n \in \mathcal{T}$.

The pair $(X, \mathcal{T})$ is a *topological space*. A set is *closed* if its complement is open.

Remark 2.3. Dually, $\varnothing$ and X are closed, arbitrary intersections of closed sets are closed, and finite unions of closed sets are closed.

Example 2.4. Let $X \neq \varnothing$ be any set.

(1) *Trivial (indiscrete) topology:* $\mathcal{T} = \{\varnothing, X\}$.

(2) *Discrete topology:* $\mathcal{T} = 2^X$.

(3) *Co-finite topology:* $A \subseteq X$ is open iff $A = \varnothing$ or $X \setminus A$ is finite.

(4) *Euclidean topology* on $\mathbb{R}^d$: the open sets of the previous section.

(5) On $X = \{0, 1\}$: the discrete topology $\{\varnothing, \{0\}, \{1\}, \{0, 1\}\}$, and also the *Sierpiński topology* $\{\varnothing, \{1\}, \{0, 1\}\}$.

2.3 Convergence and continuity

Definition 2.5. A sequence $x_1, x_2, \dots$ in $(X, \mathcal{T})$ *converges* to $x \in X$, written $x_n \to x$, if for every open $U \ni x$ there is n_0 such that $x_n \in U$ for all $n \geq n_0$.

Definition 2.6. Let $(X, \mathcal{T})$ and $(Y, \mathcal{T}')$ be topological spaces. A map $f\colon X \to Y$ is *continuous* if $f^{-1}(U) \in \mathcal{T}$ for every $U \in \mathcal{T}'$.

Exercise 2.7. Let X carry the discrete topology. Show that $f\colon \mathbb{R}^d \to X$ is continuous if and only if f is constant. *(Hint: each fibre $f^{-1}(\{x\})$ would be clopen; use that $\mathbb{R}^d$ is connected, Chapter 13.)*

Chapter 3

Bases and Sub-bases

Reading: Munkres, *Topology* (2nd ed.), §13.

3.1 Bases

Definition 3.1 (Basis)**.** A *basis* on a set X is a family $\mathcal{B} \subseteq 2^X$ such that

(B1) $\bigcup_{A\in\mathcal{B}} A = X$;

(B2) if $A_1, A_2 \in \mathcal{B}$ and $x \in A_1 \cap A_2$, there is $A_3 \in \mathcal{B}$ with $x \in A_3 \subseteq A_1 \cap A_2$.

The *topology generated by* $\mathcal{B}$ is

$$\mathcal{T}(\mathcal{B}) = \left\{\bigcup_{A\in\Gamma} A : \Gamma \subseteq \mathcal{B}\right\} = \{U \subseteq X : \forall x \in U\ \exists A \in \mathcal{B},\ x \in A \subseteq U\}.$$

Why the two descriptions agree. A union of members of $\mathcal{B}$ clearly has the local property. Conversely, if U has the local property, choose $A_x \in \mathcal{B}$ with $x \in A_x \subseteq U$ for each $x \in U$; then $U = \bigcup_{x\in U} A_x$. □

Proposition 3.2. *If $\mathcal{B}$ is a basis, then $\mathcal{T}(\mathcal{B})$ is a topology.*

Proof. $\varnothing$ is the empty union and $X \in \mathcal{T}(\mathcal{B})$ by (B1). Unions of unions of basis elements are unions of basis elements. For intersections it suffices to treat two sets $U, V \in \mathcal{T}(\mathcal{B})$: if $x \in U \cap V$, pick $A_1, A_2 \in \mathcal{B}$ with $x \in A_1 \subseteq U$, $x \in A_2 \subseteq V$, and then by (B2) some $A_3 \in \mathcal{B}$ with $x \in A_3 \subseteq A_1 \cap A_2 \subseteq U \cap V$. So $U \cap V$ has the local property. □

Proposition 3.3. *Let $\mathcal{B} \subseteq 2^X$ and suppose $\mathcal{T} = \{\bigcup_{A\in\Gamma} A : \Gamma \subseteq \mathcal{B}\}$ is a topology. Then $\mathcal{B}$ is a basis.*

Proof. (B1): $X \in \mathcal{T}$, so X is a union of members of $\mathcal{B}$. (B2): if $A_1, A_2 \in \mathcal{B}$ then $A_1 \cap A_2 \in \mathcal{T}$, so it is a union of members of $\mathcal{B}$; the member containing x is the required A_3. □

Definition 3.4. If $\mathcal{T}_1 \subseteq \mathcal{T}_2$ are topologies on X, we say $\mathcal{T}_2$ is *finer* than $\mathcal{T}_1$ (and $\mathcal{T}_1$ is *coarser* than $\mathcal{T}_2$).

Proposition 3.5 (Comparing bases)**.** *Let $\mathcal{B}_1, \mathcal{B}_2$ be bases on X. Then $\mathcal{T}(\mathcal{B}_1) \subseteq \mathcal{T}(\mathcal{B}_2)$ if and only if*

$$\forall x \in X\ \ \forall A_1 \in \mathcal{B}_1 \text{ with } x \in A_1\ \ \exists A_2 \in \mathcal{B}_2 :\ x \in A_2 \subseteq A_1.$$

Proof. ($\Rightarrow$) $A_1 \in \mathcal{T}(\mathcal{B}_1) \subseteq \mathcal{T}(\mathcal{B}_2)$, so by the local description of $\mathcal{T}(\mathcal{B}_2)$ there is such an A_2. ($\Leftarrow$) The condition says every $A_1 \in \mathcal{B}_1$ has the local property with respect to $\mathcal{B}_2$, so $\mathcal{B}_1 \subseteq \mathcal{T}(\mathcal{B}_2)$; hence all unions of members of $\mathcal{B}_1$ lie in $\mathcal{T}(\mathcal{B}_2)$. □

Example 3.6. On $\mathbb{R}^2$, both the open balls and the open rectangles $(a, b) \times (c, d)$ form bases, and

they generate the same (Euclidean) topology: every ball around x contains a rectangle around x and vice versa.

3.2 Generated topologies and sub-bases

Remark 3.7. Any intersection $\bigcap_{i\in I}\mathcal{T}_i$ of topologies on X is again a topology. Hence for an *arbitrary* family $\Gamma \subseteq 2^X$ there is a smallest topology containing Γ:

$$\mathcal{T}(\Gamma) = \bigcap\{\widehat{\mathcal{T}} : \widehat{\mathcal{T}} \text{ is a topology on } X \text{ and } \Gamma \subseteq \widehat{\mathcal{T}}\}.$$

Lemma 3.8. *If $\mathcal{B}$ is a basis, the two meanings of $\mathcal{T}(\mathcal{B})$ agree: the family of unions of members of $\mathcal{B}$ equals the smallest topology containing $\mathcal{B}$.*

Proof. Let $\mathcal{T}_1$ be the family of unions and $\mathcal{T}_2$ the smallest topology containing $\mathcal{B}$. By Theorem 3.2, $\mathcal{T}_1$ is a topology containing $\mathcal{B}$, so $\mathcal{T}_2 \subseteq \mathcal{T}_1$. Conversely, every member of $\mathcal{T}_1$ is a union of members of $\mathcal{B} \subseteq \mathcal{T}_2$, hence lies in $\mathcal{T}_2$. So $\mathcal{T}_1 = \mathcal{T}_2$. □

Definition 3.9 (Sub-basis)**.** A family $\mathcal{S} \subseteq 2^X$ with $\bigcup_{A\in\mathcal{S}} A = X$ is called a *sub-basis*. The topology it generates is $\mathcal{T}(\mathcal{S})$ as above.

Lemma 3.10 (Finite intersections)**.** *Let $\mathcal{S} \subseteq 2^X$ be a sub-basis. Then*

$$\mathcal{B}(\mathcal{S}) = \{A_1 \cap A_2 \cap \cdots \cap A_n : \ n \geq 1, \ A_i \in \mathcal{S}\}$$

is a basis.

Proof. (B1) holds since $\mathcal{S} \subseteq \mathcal{B}(\mathcal{S})$ already covers X. (B2) holds because the intersection of two finite intersections is a finite intersection, so we may take $A_3 = A_1 \cap A_2$. □

Proposition 3.11. *Let $\mathcal{S}$ be a sub-basis on X. Then $\mathcal{T}(\mathcal{S}) = \mathcal{T}\big(\mathcal{B}(\mathcal{S})\big)$. In words,*

$$\mathcal{T}(\mathcal{S}) = \{\textit{arbitrary unions of finite intersections of members of } \mathcal{S}\}.$$

Proof. $\mathcal{T}(\mathcal{B}(\mathcal{S}))$ is a topology (Theorems 3.2 and 3.10) containing $\mathcal{S}$, so $\mathcal{T}(\mathcal{S}) \subseteq \mathcal{T}(\mathcal{B}(\mathcal{S}))$. Conversely, any topology containing $\mathcal{S}$ contains all finite intersections and all unions of those, so $\mathcal{T}(\mathcal{B}(\mathcal{S})) \subseteq \mathcal{T}(\mathcal{S})$. □

Chapter 4

The Order Topology

Reading: Munkres, *Topology* (2nd ed.), §14; the dictionary order appears in §3 and §14.

Definition 4.1 (Order topology)**.** Let $(X, \leq)$ be a totally ordered set. For $a < b$ write

$$(a, b) = \{x : a < x < b\}, \quad [a, b) = \{x : a \leq x < b\}, \quad (a, b] = \{x : a < x \leq b\}.$$

The *order topology* on X is generated by the basis consisting of

- all open intervals (a, b) with $a < b$;
- all $[m, b)$ where m is the minimum of X (if it exists);
- all $(a, M]$ where M is the maximum of X (if it exists).

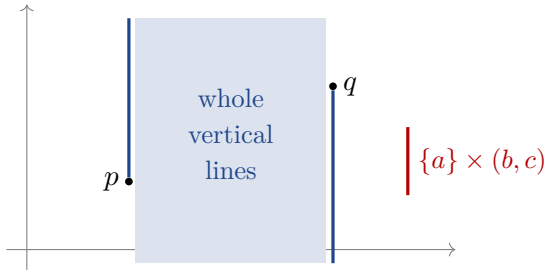
Equivalently, it is generated by the sub-basis of *open rays* $\{x : x < b\}$ and $\{x : x > a\}$, since (a, b) is the intersection of two rays.

Example 4.2. The order topology on $\mathbb{R}$ is the Euclidean topology.

Definition 4.3 (Lexicographic order)**.** The *lexicographic order* on $\mathbb{R}^2$ is the total order

$$(a, b) < (c, d) \iff a < c \text{ or } (a = c \text{ and } b < d).$$

The resulting order topology is the *lexicographic topology* on $\mathbb{R}^2$.



the interval (p, q) in the lexicographic order

Figure 4.1: An open interval in the lexicographic order. Small vertical segments $\{a\} \times (b, c) = ((a, b), (a, c))$ are already open intervals.

Proposition 4.4. *On $\mathbb{R}^d$ we have the chain*

$$\textit{discrete} \supseteq \textit{Euclidean} \supseteq \textit{co-finite} \supseteq \textit{trivial}.$$

For the lexicographic topology $\mathcal{T}_{\text{lex}}$ on $\mathbb{R}^2$:

$$\textit{discrete} \supsetneq \mathcal{T}_{\text{lex}} \supsetneq \textit{Euclidean}.$$

Proof of the statement about $\mathcal{T}_{\text{lex}}$. Every Euclidean open set U is a union of vertical segments $\{a\} \times (b, c)$ (around each point of U take a small open ball and the vertical segment through the point inside it), and these segments are lexicographic open intervals. So $\mathcal{T}_{\text{Eucl}} \subseteq \mathcal{T}_{\text{lex}}$. The vertical segment $\{0\} \times (0, 1)$ is lexicographically open but not Euclidean open, and singletons are not lexicographically open (every interval containing (a, b) contains a vertical segment), so both inclusions are strict. □

Remark 4.5. In fact $(\mathbb{R}^2, \mathcal{T}_{\text{lex}}) \cong \mathbb{R}_{\text{disc}} \times \mathbb{R}$: it is a disjoint union of uncountably many copies of $\mathbb{R}$, one for each vertical line.

Chapter 5

Products, Subspaces, Convergence and Continuity

Reading: Munkres, *Topology* (2nd ed.), §§15, 16, 19.

5.1 The product topology

Definition 5.1. Let $(X, \mathcal{T}_X)$, $(Y, \mathcal{T}_Y)$ be topological spaces. The *product topology* on $X \times Y$ is generated by the basis

$$\mathcal{B} = \{U \times V : \ U \in \mathcal{T}_X, \ V \in \mathcal{T}_Y\}.$$

($\mathcal{B}$ is a basis since $(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2)$.)

Proposition 5.2. *If $\mathcal{B}_X$ and $\mathcal{B}_Y$ are bases for $\mathcal{T}_X$ and $\mathcal{T}_Y$, then $\{A \times B : A \in \mathcal{B}_X, \ B \in \mathcal{B}_Y\}$ is a basis for the product topology.*

Proposition 5.3. *The family*

$$\mathcal{S} = \{U \times Y : U \in \mathcal{T}_X\} \cup \{X \times V : V \in \mathcal{T}_Y\} = \{\pi_X^{-1}(U)\} \cup \{\pi_Y^{-1}(V)\}$$

is a sub-basis for the product topology, since $U \times V = (U \times Y) \cap (X \times V)$.

Definition 5.4 (Arbitrary products)**.** For a family $(X_i)_{i \in I}$, the *product topology* on $X = \prod_{i \in I} X_i$ is generated by the sub-basis

$$\mathcal{S} = \{\pi_i^{-1}(U) : \ i \in I, \ U \subseteq X_i \text{ open}\}.$$

So a basic open set is $\bigcap_{k \in J} \pi_k^{-1}(U_k)$ with $J \subseteq I$ *finite*: it restricts only finitely many coordinates.

5.2 Convergence and continuity via (sub-)bases

Proposition 5.5. *Let $\mathcal{B}$ be a basis and $\mathcal{S}$ a sub-basis for $(X, \mathcal{T})$. The following are equivalent:*

(1) *$x_n \to x$;*

(2) *for every $A \in \mathcal{B}$ with $x \in A$, eventually $x_n \in A$;*

(3) *for every $A \in \mathcal{S}$ with $x \in A$, eventually $x_n \in A$.*

Proof. (1)⇒(3) and (1)⇒(2) are trivial. (2)⇒(1): an open $U \ni x$ contains some basic $A \ni x$. (3)⇒(1): an open $U \ni x$ contains some $A_1 \cap \cdots \cap A_k \ni x$ with $A_j \in \mathcal{S}$; take n_0 larger than the k

thresholds. □

Corollary 5.6. *In a product $\prod_i X_i$, $x^{(n)} \to x$ if and only if $x_i^{(n)} \to x_i$ for every i (coordinatewise convergence).*

Proposition 5.7. *Let $f\colon X \to Y$, and let $\mathcal{B}_Y$ be a basis and $\mathcal{S}_Y$ a sub-basis for $\mathcal{T}_Y$. Then*

$$f \text{ continuous} \iff \forall B \in \mathcal{B}_Y :\ f^{-1}(B) \text{ open} \iff \forall A \in \mathcal{S}_Y :\ f^{-1}(A) \text{ open}.$$

Proof. Preimages commute with unions and intersections: $f^{-1}(\bigcup_i \bigcap_{j\le k_i} A_{ij}) = \bigcup_i \bigcap_{j\le k_i} f^{-1}(A_{ij})$. □

Corollary 5.8. *A map $f\colon Z \to \prod_i X_i$ is continuous iff every coordinate $\pi_i \circ f$ is continuous.*

5.3 Subspaces

Definition 5.9. Let $(X, \mathcal{T})$ be a space and $Y \subseteq X$. The *subspace (relative) topology* on Y is

$$\mathcal{T}_Y = \{A \cap Y : A \in \mathcal{T}\}.$$

Example 5.10. (1) In $\mathbb{R}$: the subspaces $[a,b]$, (a,b), $[a,b)$. Note $[a,c)$ is open in $[a,b]$ for $a < c < b$, though not open in $\mathbb{R}$.

(2) In $\mathbb{R}^2$: the rectangle $[a,b]\times[c,d]$ and the circle $S^1 = \{(x,y) : x^2+y^2 = 1\}$.

(3) In $\mathbb{R}^{d+1}$: the sphere $S^d = \{x \in \mathbb{R}^{d+1} : \|x\| = 1\}$.

Proposition 5.11. *Let $X' \subseteq X$ and $Y' \subseteq Y$ carry the subspace topologies. On $X' \times Y'$ the following two topologies coincide:*

(1) *the product topology of the subspaces X' and Y';*

(2) *the subspace topology inherited from $X \times Y$.*

Proof. Both are generated by the same basis, because

$$(U \cap X') \times (V \cap Y') = (U \times V) \cap (X' \times Y') \qquad (U \in \mathcal{T}_X,\ V \in \mathcal{T}_Y).$$ □

Chapter 6

Closure, Interior and Boundary

Reading: Munkres, *Topology* (2nd ed.), §§17, 18.

6.1 Definitions

Definition 6.1. Let X be a space and $E \subseteq X$.

- The *closure* of E is $\mathrm{cl}(E) = \{x \in X : \ N \cap E \neq \varnothing \text{ for every open } N \ni x\}$.
- The *interior* of E is $\mathrm{int}(E) = \{x \in X : \ \exists \text{ open } N \ni x \text{ with } N \subseteq E\}$.
- The *boundary* of E is $\partial E = \mathrm{cl}(E) \setminus \mathrm{int}(E)$.

Proposition 6.2. (1) $\mathrm{cl}(E) = \bigcap\{C : \ C \text{ closed}, \ E \subseteq C\}$ *is the smallest closed set containing* E.

(2) $\mathrm{int}(E) = \bigcup\{U : \ U \text{ open}, \ U \subseteq E\}$ *is the largest open set contained in* E.

(3) $\partial E = \mathrm{cl}(E) \cap \mathrm{cl}(X \setminus E)$; *i.e.* $x \in \partial E$ *iff every open* $N \ni x$ *meets both* E *and* $X \setminus E$.

Proof. (1) $x \notin \mathrm{cl}(E)$ iff some open $N \ni x$ misses E iff x lies in some closed $C = X \setminus N \supseteq E$ with $x \notin C$. (2) is immediate from the definition. (3) follows from $X \setminus \mathrm{int}(E) = \mathrm{cl}(X \setminus E)$, which is item (8) below. □

6.2 Basic properties

Proposition 6.3. *For* $E, A, B \subseteq X$*:*

(1) $\mathrm{int}(E) \subseteq E \subseteq \mathrm{cl}(E)$;

(2) $\mathrm{cl}(\mathrm{cl}(E)) = \mathrm{cl}(E)$;

(3) $A \subseteq B \implies \mathrm{cl}(A) \subseteq \mathrm{cl}(B)$;

(4) $\mathrm{cl}(A \cup B) = \mathrm{cl}(A) \cup \mathrm{cl}(B)$;

(5) $\mathrm{cl}(A \cap B) \subseteq \mathrm{cl}(A) \cap \mathrm{cl}(B)$, *with equality not always;*

(6) $\mathrm{cl}(\varnothing) = \varnothing$;

(7) E *is closed iff* $\mathrm{cl}(E) = E$;

(8) $X \setminus \mathrm{int}(E) = \mathrm{cl}(X \setminus E)$;

(9) $X \setminus \mathrm{cl}(E) = \mathrm{int}(X \setminus E)$;

(10) *in general* $\mathrm{cl}\big(\bigcup_{i=1}^{\infty} A_i\big) \neq \bigcup_{i=1}^{\infty} \mathrm{cl}(A_i)$.

Example 6.4. For (5): in $\mathbb{R}$, $A = \mathbb{Q}$, $B = \mathbb{R} \setminus \mathbb{Q}$ gives $\mathrm{cl}(A \cap B) = \varnothing$ but $\mathrm{cl}(A) \cap \mathrm{cl}(B) = \mathbb{R}$. For (10): with $A_i = \{1/i\}$, the union of closures is $\{1/i : i \in \mathbb{N}\}$, while the closure of the union also contains 0.

Proposition 6.5. *For $E \subseteq X$:*

(1) $\mathrm{int}(E) \cap \partial E = \varnothing$;

(2) $\partial E = \varnothing$ *iff* E *is clopen;*

(3) $\mathrm{cl}(E) = E \cup \partial E = \mathrm{int}(E) \cup \partial E$;

(4) $\partial E = \partial(X \setminus E)$, *and* ∂E *is closed, so* $\partial(\partial E) \subseteq \partial E$.

6.3 Isolated points and limit points

Definition 6.6. Let $E \subseteq X$.

(1) $x \in E$ is an *isolated point* of E if there is an open $N \ni x$ with $N \cap E = \{x\}$.

(2) $x \in X$ is a *limit point* (accumulation point) of E if every open $N \ni x$ satisfies $(N \setminus \{x\}) \cap E \neq \varnothing$.

We write $\mathrm{iso}(E)$ and $\lim(E)$ for the sets of isolated and limit points.

Proposition 6.7. *For any $E \subseteq X$, $\mathrm{cl}(E) = \mathrm{iso}(E) \sqcup \lim(E) = E \cup \lim(E)$.*

Proof. Let $x \in \mathrm{cl}(E)$. If x is not a limit point, some open $N \ni x$ has $(N \setminus \{x\}) \cap E = \varnothing$; since $N \cap E \neq \varnothing$, we get $N \cap E = \{x\}$, so $x \in \mathrm{iso}(E)$. The union is disjoint, and the reverse inclusions are clear. □

Proposition 6.8. (1) *If x is the limit of a sequence in E, then $x \in \mathrm{cl}(E)$.*

(2) *If x is the limit of a sequence in $E \setminus \{x\}$, then $x \in \lim(E)$.*

Proof. If $x_n \to x$ and $N \ni x$ is open, then $x_n \in N$ for large n, so N meets E (resp. $E \setminus \{x\}$). □

Remark 6.9. The converses fail in general (see Theorem 14.20); they hold in metrizable spaces (Theorem 9.12).

6.4 Characterisations of continuity

Proposition 6.10. *For $f\colon X \to Y$ the following are equivalent:*

(1) *$f^{-1}(U)$ is open for every open $U \subseteq Y$;*

(2) *$f^{-1}(V)$ is closed for every closed $V \subseteq Y$;*

(3) *$f(\mathrm{cl}(E)) \subseteq \mathrm{cl}(f(E))$ for every $E \subseteq X$;*

(4) *f is continuous at every $x \in X$: for every open $B \ni f(x)$ there is an open $A \ni x$ with $f(A) \subseteq B$.*

Proof. (1)⇔(2): $f^{-1}(Y \setminus V) = X \setminus f^{-1}(V)$.

(2)⇒(3): $V = \mathrm{cl}(f(E))$ is closed, so $f^{-1}(V)$ is a closed set containing E; hence $\mathrm{cl}(E) \subseteq f^{-1}(V)$ and $f(\mathrm{cl}(E)) \subseteq V$.

(3)⇒(2): Let V be closed and $U = f^{-1}(V)$. Then $f(\mathrm{cl}(U)) \subseteq \mathrm{cl}(f(U)) \subseteq \mathrm{cl}(V) = V$, so $\mathrm{cl}(U) \subseteq f^{-1}(V) = U$ and U is closed.

(1)⇒(4): take $A = f^{-1}(B)$.

(4)⇒(1): if U is open and $x \in f^{-1}(U)$, there is an open $A_x \ni x$ with $A_x \subseteq f^{-1}(U)$; so $f^{-1}(U) = \bigcup_x A_x$ is open. □

Proposition 6.11 (Continuity preserves limits)**.** *If $f\colon X \to Y$ is continuous and $x_n \to x$, then $f(x_n) \to f(x)$.*

Proof. If $V \ni f(x)$ is open, then $f^{-1}(V) \ni x$ is open, so eventually $x_n \in f^{-1}(V)$, i.e. $f(x_n) \in V$. □

Chapter 7

Homeomorphisms and Hausdorff Spaces

Reading: Munkres, *Topology* (2nd ed.), §§17, 18.

7.1 Homeomorphisms and embeddings

Definition 7.1. A map $f\colon X \to Y$ is a *homeomorphism* if f is a bijection and both f and f^{-1} are continuous. Then X and Y are *homeomorphic* (*topologically equivalent*), written $X \cong Y$.

Remark 7.2. A homeomorphism is exactly a bijection f with U open $\iff$ $f(U)$ open. So X and Y are "the same space" with points renamed.

Definition 7.3 (Embedding)**.** An injective continuous map $f\colon X \to Y$ is an *embedding* if $f\colon X \to f(X)$ is a homeomorphism, where $f(X)$ carries the subspace topology.

Example 7.4. $t \mapsto (\cos t, \sin t)$ from $[0, 2\pi)$ to $\mathbb{R}^2$ is a continuous injection but *not* an embedding: points near 2π map close to the image of 0, so the inverse $S^1 \to [0, 2\pi)$ is not continuous at $(1, 0)$.

7.2 Hausdorff spaces

Definition 7.5. A space X is *Hausdorff* if for all distinct $a, b \in X$ there are open $A \ni a$, $B \ni b$ with $A \cap B = \varnothing$.

Proposition 7.6. *If X is Hausdorff, then:*

(1) *every singleton $\{x\}$ is closed;*

(2) *every finite subset of X is closed;*

(3) *a sequence in X has at most one limit.*

Proof. (1) For $y \neq x$ choose disjoint open $A \ni x$, $B \ni y$; then $B \subseteq X \setminus \{x\}$, so $X \setminus \{x\}$ is open. (2) Finite unions of closed sets are closed. (3) If $x_n \to a$ and $x_n \to b$ with $a \neq b$, take disjoint $A \ni a$, $B \ni b$; eventually $x_n \in A \cap B = \varnothing$, which is absurd. □

Example 7.7. In an infinite set with the co-finite topology, any two non-empty open sets intersect, so it is not Hausdorff; indeed the sequence $1, 2, 3, \ldots$ in $\mathbb{N}$ converges to *every* point.

Chapter 8

Topological Properties and More on Continuity

Reading: Munkres, *Topology* (2nd ed.), §18.

8.1 Topological invariants

Definition 8.1. A property P of spaces is a *topological property* (*invariant*) if whenever $X \cong Y$ and X has P, then Y has P.

Example 8.2. Cardinality, being Hausdorff, connectedness, path-connectedness, compactness, metrizability, the fundamental group, and topological dimension are all topological invariants. Boundedness is *not*: $(0,1) \cong \mathbb{R}$.

8.2 Building continuous maps

Proposition 8.3. (1) *Constant maps are continuous.*

(2) *The inclusion $i\colon X_0 \hookrightarrow X$ of a subspace is continuous.*

(3) *If $f\colon X \to Y$ and $g\colon Y \to Z$ are continuous, so is $g \circ f$.*

(4) *If $f\colon X \to Y$ is continuous and $X_0 \subseteq X$, then $f|_{X_0} = f \circ i$ is continuous.*

(5) *The projections $\pi_X\colon X \times Y \to X$, $\pi_Y\colon X \times Y \to Y$ are continuous.*

Proof. (2) $i^{-1}(U) = U \cap X_0$. (3) $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W))$. (5) $\pi_X^{-1}(U) = U \times Y$. □

Proposition 8.4 (Pasting lemma)**.** *Let $X = A \cup B$ where A, B are both closed (or both open) in X. Let $f\colon A \to Y$ and $g\colon B \to Y$ be continuous with $f = g$ on $A \cap B$. Then $h\colon X \to Y$, $h|_A = f$, $h|_B = g$, is continuous.*

Proof. Closed case: for $V \subseteq Y$ closed, $h^{-1}(V) = f^{-1}(V) \cup g^{-1}(V)$. Now $f^{-1}(V)$ is closed in A, hence closed in X since A is closed in X; likewise $g^{-1}(V)$. A union of two closed sets is closed. The open case is identical with open sets. □

Remark 8.5. The pasting lemma fails for arbitrary pieces. The Dirichlet function $\mathbf{1}_{\mathbb{Q}}\colon \mathbb{R} \to \mathbb{R}$ is constant (hence continuous) on $\mathbb{Q}$ and on $\mathbb{R} \setminus \mathbb{Q}$, yet continuous nowhere.

8.3 Preview: Urysohn and Tietze

Two questions will guide us later.

Question 8.6 (Urysohn)**.** Given disjoint $A, B \subseteq X$, is there a continuous $f\colon X \to [0,1]$ with $f(A) = \{0\}$ and $f(B) = \{1\}$?

This is answered for closed A, B in *normal* spaces by Urysohn's lemma (Theorem 16.1).

Question 8.7 (Tietze)**.** Let $A \subseteq X$ be closed and $f\colon A \to \mathbb{R}$ continuous. Is there a continuous $g\colon X \to \mathbb{R}$ with $g|_A = f$?

Again yes for normal X (Theorem 16.3).

Remark 8.8. Closedness of A matters. A continuous $f\colon \mathbb{Q} \to \mathbb{R}$ extends continuously to $\mathbb{R}$ if and only if f is uniformly continuous on $\mathbb{Q} \cap [-n, n]$ for every n; e.g. $f(x) = \frac{1}{x-\sqrt{2}}$ is continuous on $\mathbb{Q}$ but has no continuous extension.

Chapter 9

Metrizable Spaces

Reading: Munkres, *Topology* (2nd ed.), §§20, 21.

9.1 Metrics and the topologies they define

Definition 9.1. A space $(X, \mathcal{T})$ is *metrizable* if there is a metric d on X whose open balls generate $\mathcal{T}$. Two metrics d_1, d_2 on X are *(topologically) equivalent* if they generate the same topology.

Remark 9.2. A *metric space* comes with a chosen metric; *metrizability* is a property of a topology that asks only that *some* compatible metric exists. Many metrics give the same topology.

Proposition 9.3. *Every metric is equivalent to a bounded metric, e.g.*

$$\bar{d}(x,y) = \min\{d(x,y), 1\} \le 1 \qquad \text{or} \qquad \bar{d}(x,y) = \frac{d(x,y)}{1+d(x,y)} < 1.$$

Idea. Both metrics have the same small balls as d: e.g. for $\varepsilon < 1$ the $\bar{d}$-ball and d-ball of radius ε coincide in the first case. Balls of small radius are enough to determine the topology. □

Example 9.4 (Discrete spaces)**.** The discrete topology on any set is metrizable via $d(x,y) = 0$ if $x = y$ and 1 otherwise; then $B_{1/2}(x) = \{x\}$.

Example 9.5 (Cantor space $\{0,1\}^{\mathbb{N}}$)**.** The product topology on $\{0,1\}^{\mathbb{N}}$ is metrizable by

$$d(x,y) = 2^{-n(x,y)}, \qquad n(x,y) = \min\{n : x_n \neq y_n\} \quad (d(x,x) = 0).$$

The ball $B_{2^{-n}}(x)$ is exactly the cylinder $\{y : y_1 = x_1, \dots, y_n = x_n\}$. An equivalent metric is

$$d'(x,y) = \sum_{n=1}^{\infty} \frac{d_0(x_n, y_n)}{2^n}, \qquad d_0 = \text{discrete metric on } \{0,1\},$$

since $2^{-n(x,y)} \le d'(x,y) \le 2 \cdot 2^{-n(x,y)}$.

9.2 A non-metrizable product

Definition 9.6 (Neighbourhood basis; first countability)**.** A *neighbourhood basis* at $x \in X$ is a family $\{N_i : i \in I\}$ of open sets containing x such that every open $U \ni x$ contains some N_i. The space X is *first countable* if every point has a countable neighbourhood basis.

Remark 9.7. Every metrizable space is first countable: $\{B_{1/n}(x) : n \in \mathbb{N}\}$ is a neighbourhood basis at x.

Example 9.8. Let I be uncountable and $\mathcal{X} = \{0,1\}^I$ with the product topology. Then $\mathcal{X}$ is not first countable, hence not metrizable.

Proof. Basic neighbourhoods of x are the cylinders

$$C(K,x) = \{y \in \mathcal{X} : \ y_i = x_i \ \forall i \in K\}, \qquad K \subseteq I \text{ finite}.$$

Suppose $\{N_n\}_{n\in\mathbb{N}}$ were a countable neighbourhood basis at x. Each N_n contains some cylinder $C(K_n,x)$, so $C(K_n,x) \subseteq N_n$. The set $\bigcup_n K_n$ is countable, so pick $j \in I \setminus \bigcup_n K_n$. The open set $U = \{y : y_j = x_j\}$ contains x, but no N_n fits inside U: the point y that agrees with x except at coordinate j lies in every $C(K_n,x) \subseteq N_n$ but not in U. Contradiction. □

Theorem 9.9. *Let $(X_i)_{i\in I}$ be metrizable spaces, each with at least two points. Then $\prod_{i\in I} X_i$ is metrizable if and only if I is countable.*

Idea. If $I = \mathbb{N}$ and d_i is a metric on X_i bounded by 1, then $d(x,y) = \sum_i 2^{-i} d_i(x_i,y_i)$ works. If I is uncountable, the product contains a copy of $\{0,1\}^I$, which is not metrizable by Theorem 9.8. □

9.3 Three topologies on $\mathbb{R}^{\mathbb{N}}$

Example 9.10. Let $\bar{d}(s,t) = \min\{|s-t|, 1\}$ on $\mathbb{R}$.

(1) **Product topology**: metrizable by $d(x,y) = \sum_{i=1}^{\infty} 2^{-i}\, \bar{d}(x_i,y_i)$.

(2) **Uniform topology**: induced by the uniform metric $\rho(x,y) = \sup_i \bar{d}(x_i,y_i)$.

(3) **Box topology**: generated by all $\prod_i U_i$ with each $U_i \subseteq \mathbb{R}$ open. It is not metrizable.

These are related by

$$\mathcal{T}_{\text{product}} \subsetneq \mathcal{T}_{\text{uniform}} \subsetneq \mathcal{T}_{\text{box}}.$$

9.4 Sequences suffice in metrizable spaces

In general spaces, sequences do not capture the topology (Theorem 14.20). In metrizable spaces they do.

Theorem 9.11. *Let X be metrizable, Y any space, $f\colon X \to Y$ and $x \in X$. Then f is continuous at x if and only if $x_n \to x$ implies $f(x_n) \to f(x)$.*

Proof. ($\Rightarrow$) As in Theorem 6.11.

($\Leftarrow$) Suppose f is not continuous at x. Then there is an open $V \ni f(x)$ such that no open $U \ni x$ satisfies $f(U) \subseteq V$. Let d be a compatible metric. For each n the ball $B_{1/n}(x)$ is not mapped into V, so choose $x_n \in B_{1/n}(x)$ with $f(x_n) \notin V$. Then $d(x_n,x) < 1/n \to 0$, so $x_n \to x$, but $f(x_n) \notin V$ for all n, so $f(x_n) \not\to f(x)$. □

Theorem 9.12. *Let X be metrizable and $E \subseteq X$. Then*

(1) $\operatorname{cl}(E) = \{x \in X : \exists (x_n) \subseteq E,\ x_n \to x\}$;

(2) $\lim(E) = \{x \in X : \exists (x_n) \subseteq E \setminus \{x\},\ x_n \to x\}$.

Proof. ($\supseteq$) holds in every space (Theorem 6.8). ($\subseteq$) Fix a compatible metric d. If $x \in \mathrm{cl}(E)$, each ball $B_{1/n}(x)$ meets E; choose $x_n \in B_{1/n}(x) \cap E$. Then $d(x_n, x) < 1/n$, so $x_n \to x$. For (2) choose $x_n \in (B_{1/n}(x) \setminus \{x\}) \cap E$ instead. □

Remark 9.13. Both proofs only used that $\{B_{1/n}(x)\}_n$ is a countable neighbourhood basis. So both theorems remain true for *first countable* spaces.

Chapter 10

Uniform Continuity and Uniform Convergence

Reading: Munkres, *Topology* (2nd ed.), §§21, 43.

10.1 Uniform continuity

Definition 10.1. A map $f\colon X \to Y$ between metric spaces is *uniformly continuous* if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, x' \in X:\ d(x,x') < \delta \implies d(f(x), f(x')) < \varepsilon.$$

The point: δ does not depend on x.

Example 10.2. $f\colon (0,1) \to \mathbb{R}$, $f(x) = 1/x$, is continuous but not uniformly continuous. Take $\varepsilon = 1$ and any $\delta > 0$. Let $x \in (0, \min\{\delta, 1\})$ and $x' = x/2$. Then $|x - x'| = x/2 < \delta$ but $|f(x') - f(x)| = \frac{2}{x} - \frac{1}{x} = \frac{1}{x} > 1$.

10.2 Convergence of functions

Let X be a set and Y a space. There are two natural notions of convergence for maps $f_n\colon X \to Y$.

Definition 10.3. (1) *Pointwise convergence*: $f_n \to f$ if $f_n(x) \to f(x)$ for every $x \in X$. This is convergence in the product topology on Y^X.

(2) *Uniform convergence* (Y metric): $f_n \to f$ uniformly if

$$\forall \varepsilon > 0\ \exists n_0\ \forall n \geq n_0\ \forall x \in X:\ d(f_n(x), f(x)) < \varepsilon.$$

This is convergence for the uniform metric $\rho(f,g) = \sup_{x \in X} \bar{d}(f(x), g(x))$ on Y^X.

Example 10.4. On $X = Y = [0,1]$ let $f_n(x) = x^n$. Then $f_n \to f$ pointwise, where $f(1) = 1$ and $f(x) = 0$ for $x < 1$. The convergence is not uniform: for every n, $f_n(x) > 1/2$ for $x \in (2^{-1/n}, 1)$, so $\sup_x |f_n(x) - f(x)| \geq 1/2$. This is also forced by the next theorem, since f is discontinuous.

Theorem 10.5 (Uniform limit theorem)**.** *Let X be a space, Y a metric space, and $f_n\colon X \to Y$ continuous. If $f_n \to f$ uniformly, then f is continuous.*

Proof. Let $x_0 \in X$ and $\varepsilon > 0$. Choose n with $d(f_n(x), f(x)) < \varepsilon/3$ for all $x \in X$. Since f_n is continuous there is an open $N \ni x_0$ with $f_n(N) \subseteq B_{\varepsilon/3}(f_n(x_0))$. For $x \in N$,

$$d(f(x), f(x_0)) \leq d(f(x), f_n(x)) + d(f_n(x), f_n(x_0)) + d(f_n(x_0), f(x_0)) < \tfrac{\varepsilon}{3} + \tfrac{\varepsilon}{3} + \tfrac{\varepsilon}{3} = \varepsilon.$$

So $f(N) \subseteq B_\varepsilon(f(x_0))$ and f is continuous at x_0. □

Remark 10.6. Two ways to read the theorem:

(1) The set $C(X, Y)$ of continuous maps is *closed* in Y^X with the uniform topology.

(2) If for every $\varepsilon > 0$ there is a continuous f_ε that ε-approximates f (i.e. $d(f(x), f_\varepsilon(x)) < \varepsilon$ for all x), then f is continuous.

Chapter 11

Gluing Constructions

Reading: Munkres, *Topology* (2nd ed.), §22.

Many spaces are built by gluing simpler ones.

Example 11.1.

- The circle: $S^1 \cong [0, 2\pi]/(0 \sim 2\pi)$ via $t \mapsto (\cos t, \sin t)$.
- The sphere: $S^2 \cong D^2/\partial D^2$, the closed disk with its boundary circle collapsed to one point. Equivalently, S^2 is two copies of D^2 glued along their boundary circles.
- A three-armed star: three copies of $[0, 1]$ with their left endpoints identified, $0 \sim 0' \sim 0''$.
- The torus: a closed square with opposite edges identified with the same orientation.
- The Klein bottle: a closed square with one pair of opposite edges identified with the same orientation and the other pair with opposite orientation.

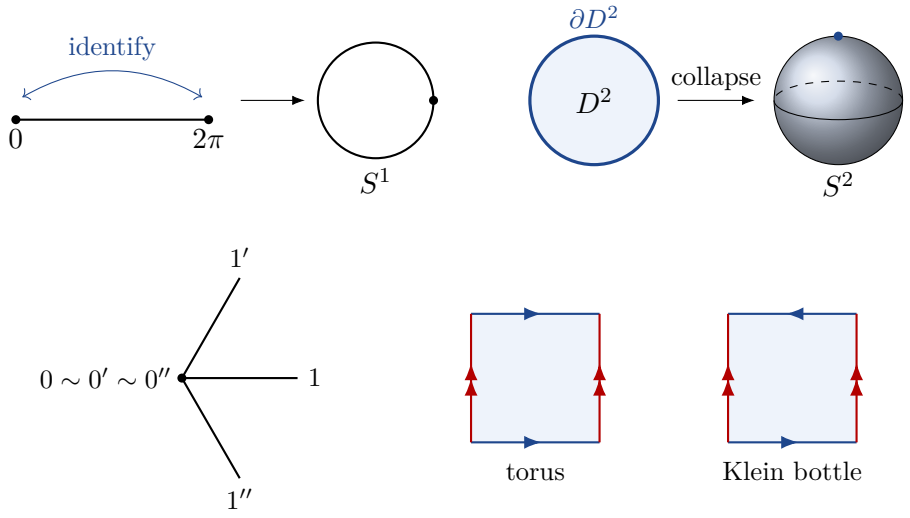


Figure 11.1: Gluing constructions: $S^1 \cong [0, 2\pi]/(0 \sim 2\pi)$; $S^2 \cong D^2/\partial D^2$; a wedge of three intervals; the torus and Klein bottle as identified squares (edges with matching arrows are glued so the arrows agree).

11.1 Two points of view

To make "gluing" precise we must say which subsets of the glued space are open. Using the circle as the model:

- *Via a quotient map.* Let $q\colon [0, 2\pi] \to S^1$, $q(t) = (\cos t, \sin t)$. Declare $B \subseteq S^1$ open iff $q^{-1}(B)$ is open in $[0, 2\pi]$.
- *Via an equivalence relation.* On $X = [0, 2\pi]$ let $a \sim b$ iff $a = b$ or $\{a, b\} = \{0, 2\pi\}$. Then $X/\!\sim\, = \{[x] : x \in X\}$ with $[0] = [2\pi]$, and $B \subseteq X/\!\sim$ is open iff $\bigcup_{[x]\in B}[x]$ is open in X.

The next chapter formalises both.

Chapter 12

Initial and Final Topologies; Quotients

Reading: Munkres, *Topology* (2nd ed.), §22.

Definition 12.1 (Initial (weak) topology)**.** Let $f\colon X \to Y$ where Y is a space. The coarsest topology on X making f continuous is

$$\mathcal{T}_{\text{init}} = \{f^{-1}(U) : U \subseteq Y \text{ open}\}.$$

This is a topology because preimages preserve arbitrary unions and finite intersections.

Example 12.2. The subspace topology is the initial topology for the inclusion $Y \hookrightarrow X$, and the product topology is the coarsest topology making all projections π_i continuous.

Definition 12.3 (Final (strong) topology)**.** Let $p\colon X \to Y$ where X is a space. The finest topology on Y making p continuous is

$$\mathcal{T}_{\text{final}} = \{B \subseteq Y : p^{-1}(B) \text{ is open in } X\}.$$

Definition 12.4 (Quotient map)**.** A surjection $q\colon X \to Y$ between spaces is a *quotient map* if for every $B \subseteq Y$,

$$B \text{ is open in } Y \iff q^{-1}(B) \text{ is open in } X,$$

i.e. Y carries the final topology for q.

Definition 12.5 (Quotient by an equivalence relation)**.** Let $\sim$ be an equivalence relation on a space X, let $X/\!\sim = \{[x] : x \in X\}$ and $q\colon X \to X/\!\sim$, $q(x) = [x]$. The *quotient topology* on $X/\!\sim$ is the final topology for q:

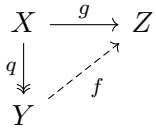
$$B \subseteq X/\!\sim \text{ is open} \iff q^{-1}(B) = \bigcup_{[x]\in B} [x] \text{ is open in } X.$$

12.1 Maps out of a quotient

Proposition 12.6. *Let $q\colon X \to Y$ be a quotient map and $f\colon Y \to Z$ any map. Then f is continuous if and only if $f \circ q$ is continuous.*

Proof. ($\Rightarrow$) A composition of continuous maps is continuous. ($\Leftarrow$) Let $C \subseteq Z$ be open. Then $(f \circ q)^{-1}(C) = q^{-1}(f^{-1}(C))$ is open in X, so $f^{-1}(C)$ is open in Y by the definition of a quotient map. □

Proposition 12.7 (Universal property)**.** *Let $q\colon X \to Y$ be a quotient map and $g\colon X \to Z$ continuous. If g is constant on each fibre $q^{-1}(\{y\})$, then there is a unique map $f\colon Y \to Z$ with $g = f \circ q$, and f is continuous.*



Proof. Define $f(y) = g(x)$ for any $x \in q^{-1}(\{y\})$; this is well defined because g is constant on fibres and q is surjective, and it is forced by $g = f \circ q$, so it is unique. It is continuous by Theorem 12.6. □

12.2 Quotients need not be Hausdorff

Example 12.8 (Line with two origins)**.** Let $X = [0,1] \times \{0,1\}$ (two copies of $[0,1]$, a subspace of $\mathbb{R}^2$), and glue the copies everywhere except at 0:

$$(x,0) \sim (x,1) \quad \text{for } x \in (0,1], \qquad (0,0) \not\sim (0,1).$$

Then X is Hausdorff but $X/\!\sim$ is not.

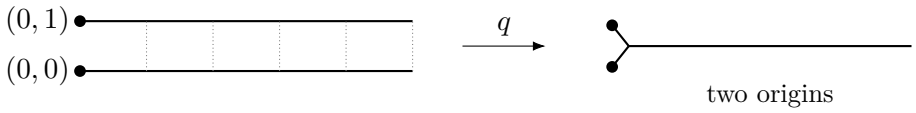


Figure 12.1: Gluing two segments everywhere except at 0.

Claim. Every open $N \ni [(0,0)]$ and every open $N' \ni [(0,1)]$ intersect.

Proof. $q^{-1}(N)$ is open in X and contains $(0,0)$, so $[0,\varepsilon) \times \{0\} \subseteq q^{-1}(N)$ for some $\varepsilon > 0$. Since $q^{-1}(N)$ is a union of equivalence classes, for $a \in (0,\varepsilon)$ we also get $(a,1) \in q^{-1}(N)$. Similarly, shrinking ε, $(a,0), (a,1) \in q^{-1}(N')$ for all $a \in (0,\varepsilon)$. Hence for any such a, $[(a,0)] \in N \cap N'$. □

Chapter 13

Connectedness

Reading: Munkres, *Topology* (2nd ed.), §§23–25.

13.1 Connected and disconnected spaces

Definition 13.1. A space X is *disconnected* if there are non-empty open sets A, B with $A \cap B = \varnothing$ and $A \cup B = X$ (a *separation* of X). Otherwise X is *connected*. A subset is connected if it is connected in the subspace topology.

Remark 13.2. In a separation, $A = X \setminus B$ and $B = X \setminus A$ are both clopen. So X is connected if and only if its only clopen subsets are $\varnothing$ and X.

Definition 13.3. X is *totally disconnected* if its only non-empty connected subsets are singletons.

Example 13.4. Connected: $\mathbb{R}$, $\mathbb{R}^d$, any interval (e.g. $[0,1]$, $(0,\infty)$), any ball in $\mathbb{R}^d$.

Disconnected: $[0,1) \cup [2,3]$; $(0,1) \cup (1,2)$; the lexicographic plane (each vertical line is clopen). **Totally disconnected:** $\{0,1\}^{\mathbb{N}}$, $\mathbb{Q}$, $\mathbb{R} \setminus \mathbb{Q}$, any discrete space, and $\mathbb{Z}$ with the *evenly spaced* (Furstenberg) topology, whose basic open sets are the arithmetic progressions $a + b\mathbb{Z}$, $b \geq 1$; these are clopen.

Proposition 13.5. *The continuous image of a connected space is connected.*

Proof. Let $f\colon X \to Y$ be continuous with X connected, and suppose $f(X) = U \cup V$ is a separation in the subspace $f(X)$. Then $f^{-1}(U)$ and $f^{-1}(V)$ are open in X (as $f\colon X \to f(X)$ is continuous), disjoint, non-empty (since $U, V \subseteq f(X)$ are non-empty), and cover X. This separates X, a contradiction. □

Proposition 13.6. *Let $\{A_i\}_{i \in I}$ be connected subsets of X with a common point $x_0 \in \bigcap_i A_i$. Then $A = \bigcup_i A_i$ is connected.*

Proof. Suppose $A = U \cup V$ with U, V disjoint and open in A. For each i, $A_i = (A_i \cap U) \cup (A_i \cap V)$ is a decomposition into disjoint relatively open sets, so by connectedness $A_i \subseteq U$ or $A_i \subseteq V$. Say $x_0 \in U$. Then every A_i meets U, hence $A_i \subseteq U$ for all i, so $A \subseteq U$ and $V = \varnothing$. Thus A has no separation. □

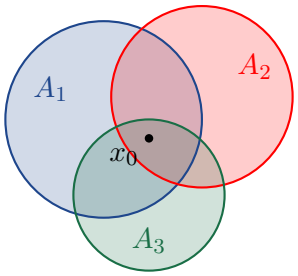


Figure 13.1: Connected sets sharing a common point x_0 have a connected union.

13.2 Mutually separated sets

Definition 13.7. $A, B \subseteq X$ are *mutually separated* if $\mathrm{cl}(A) \cap B = \varnothing = A \cap \mathrm{cl}(B)$. (In particular $A \cap B = \varnothing$.)

Proposition 13.8. *If $A, B \subseteq X$ are mutually separated and $A \cup B = X$, then A and B are clopen.*

Proof. $A \cap B \subseteq A \cap \mathrm{cl}(B) = \varnothing$, so $B = X \setminus A$ and $A = X \setminus B$. From $\mathrm{cl}(A) \cap B = \varnothing$ we get $\mathrm{cl}(A) \subseteq X \setminus B = A$, so A is closed; symmetrically B is closed. Each is the complement of the other, so both are also open. □

Proposition 13.9. *A subset $Y \subseteq X$ is disconnected if and only if there are mutually separated $A, B \subseteq X$ with $Y \subseteq A \cup B$, $A \cap Y \neq \varnothing$ and $B \cap Y \neq \varnothing$.*

Proof. ($\Rightarrow$) Let $Y = C \cup D$ be a separation in Y. Then C is closed in Y, so $\mathrm{cl}_X(C) \cap Y = \mathrm{cl}_Y(C) = C$ and hence $\mathrm{cl}_X(C) \cap D = \mathrm{cl}_X(C) \cap Y \cap D = C \cap D = \varnothing$. Symmetrically $C \cap \mathrm{cl}_X(D) = \varnothing$. Take $A = C$, $B = D$.

($\Leftarrow$) Put $U = A \cap Y$ and $V = B \cap Y$: non-empty, disjoint, with $U \cup V = Y$. Moreover $\mathrm{cl}_Y(U) \cap V \subseteq \mathrm{cl}_X(A) \cap B = \varnothing$, so $\mathrm{cl}_Y(U) \subseteq Y \setminus V = U$ and U is closed in Y; likewise V. As complements of each other in Y, both are open in Y, giving a separation. □

Proposition 13.10. *Let $A, B \subseteq X$ be mutually separated and $E \subseteq A \cup B$ connected. Then $E \subseteq A$ or $E \subseteq B$.*

Proof. If E met both A and B, Theorem 13.9 would make E disconnected. □

Proposition 13.11. *If $E \subseteq X$ is connected and $E \subseteq F \subseteq \mathrm{cl}(E)$, then F is connected. In particular $\mathrm{cl}(E)$ is connected.*

Proof. Suppose $F = U \cup V$ is a separation in F. Since E is connected, $E \cap U$ and $E \cap V$ cannot both be non-empty (they would separate E), so say $E \subseteq U$. Write $V = O \cap F$ with O open in X and pick $y \in V$. Since $y \in \mathrm{cl}(E)$, the open set $O \ni y$ meets E; but $O \cap E = O \cap F \cap E = V \cap E \subseteq V \cap U = \varnothing$. Contradiction. □

13.3 Products of connected spaces

Theorem 13.12. *Let $(X_i)_{i\in I}$ be non-empty spaces. Then $\prod_{i\in I} X_i$ (product topology) is connected if and only if every X_i is connected.*

Proof of ($\Rightarrow$). Each $X_i = \pi_i(\prod_j X_j)$ is a continuous image of a connected space. □

Proof of ($\Leftarrow$) for two factors. Let X, Y be connected and suppose $X \times Y = A \cup B$ with A, B non-empty, disjoint and open.

Claim 1. Some horizontal slice $X \times \{y_0\}$ or some vertical slice $\{x_0\} \times Y$ meets both A and B.

Proof. Take $(x_1, y_1) \in A$ and $(x_2, y_2) \in B$ and look at the corner (x_1, y_2). If $(x_1, y_2) \in A$, the slice $X \times \{y_2\}$ contains $(x_1, y_2) \in A$ and $(x_2, y_2) \in B$. If $(x_1, y_2) \in B$, the slice $\{x_1\} \times Y$ contains $(x_1, y_1) \in A$ and $(x_1, y_2) \in B$. □

Say $X \times \{y_0\}$ meets both (the vertical case is symmetric), and put

$$A_{y_0} = \{x \in X : (x, y_0) \in A\}, \qquad B_{y_0} = \{x \in X : (x, y_0) \in B\}.$$

Claim 2. A_{y_0}, B_{y_0} are non-empty, disjoint, open in X, and $A_{y_0} \cup B_{y_0} = X$.

Proof. Only openness needs an argument. If $x \in A_{y_0}$ then $(x, y_0) \in A$, and since A is open there are open $V \subseteq X$, $W \subseteq Y$ with $(x, y_0) \in V \times W \subseteq A$. Then $x \in V \subseteq A_{y_0}$. Likewise for B_{y_0}. □

Thus X is disconnected, a contradiction. □

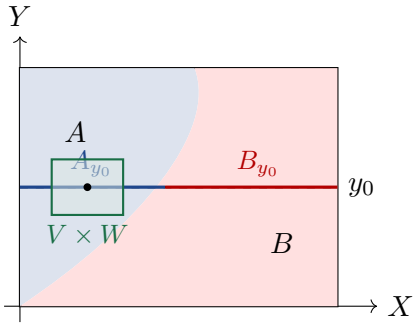


Figure 13.2: A hypothetical separation $A \cup B$ of $X \times Y$ would cut a slice $X \times \{y_0\} \cong X$ into two open pieces.

Proof of ($\Leftarrow$), general case. By induction, finite products of connected spaces are connected. Fix $a \in X = \prod_i X_i$. For finite $J \subseteq I$ let

$$\tilde{X}_J = \{x \in X : \ x_i = a_i \text{ for all } i \notin J\} \cong \prod_{i\in J} X_i,$$

which is connected. All $\tilde{X}_J$ contain a, so $D = \bigcup_{J \text{ finite}} \tilde{X}_J$ is connected by Theorem 13.6.

Claim 3. D is dense in X.

Proof. A non-empty basic open set has the form $B = \bigcap_{k\in J} \pi_k^{-1}(U_k)$ with J finite. Pick $b \in B$ and define $x_i = b_i$ for $i \in J$ and $x_i = a_i$ for $i \notin J$. Then $x \in B \cap \tilde{X}_J \subseteq B \cap D$. □

Hence $X = \mathrm{cl}(D)$ is connected by Theorem 13.11. □

13.4 Connected components

For $x, y \in X$ write $x \sim y$ if some connected $C \subseteq X$ contains both x and y. This is an equivalence relation: reflexivity uses $C = \{x\}$, and transitivity holds because if $C \ni x, y$ and $C' \ni y, z$ are connected then $C \cup C'$ is connected (Theorem 13.6).

Definition 13.13. The equivalence classes of $\sim$ are the *connected components* of X. The component of x is denoted $C(x)$.

Proposition 13.14. (1) *Each component is connected; $C(x)$ is the largest connected subset of X containing x.*

(2) *Each component is closed.*

Proof. (1) $C(x)$ is the union of all connected sets containing x, which is connected by Theorem 13.6, and it contains every connected set through x. (2) $\mathrm{cl}(C(x))$ is connected (Theorem 13.11) and contains x, so $\mathrm{cl}(C(x)) \subseteq C(x)$ by maximality. □

Example 13.15. The components of $(0,1) \cup (1,2)$ are $(0,1)$ and $(1,2)$. The components of $\mathbb{Q}$ are the singletons. Components need not be open: in $\mathbb{Q}$, singletons are not open.

13.5 Connected subsets of $\mathbb{R}$

Proposition 13.16. *A set $I \subseteq \mathbb{R}$ is connected if and only if it is an interval (i.e. $a, b \in I$ and $a < c < b$ imply $c \in I$).*

Proof. ($\Rightarrow$) If $a, b \in I$, $a < c < b$ and $c \notin I$, then $I \cap (-\infty, c)$ and $I \cap (c, \infty)$ separate I.

($\Leftarrow$) Let I be an interval and suppose $I = A \cup B$ with A, B non-empty, disjoint and open in I. Pick $a \in A$, $b \in B$; WLOG $a < b$, so $[a, b] \subseteq I$. Let $c = \sup(A \cap [a, b]) \in [a, b] \subseteq I$.

- If $c \in A$: then $c \neq b$, so $c < b$. As A is open in I there is $\varepsilon > 0$ with $[c, c+\varepsilon) \subseteq A$ and $c+\varepsilon \leq b$, giving points of $A \cap [a, b]$ above c. Contradiction.
- If $c \in B$: then $c \neq a$, so $c > a$. As B is open in I there is $\varepsilon > 0$ with $(c-\varepsilon, c] \subseteq B$ and $c - \varepsilon \geq a$. Then $A \cap [a, b] \subseteq [a, c-\varepsilon]$, so $c \leq c - \varepsilon$. Contradiction. □

Theorem 13.17 (Intermediate value theorem)**.** *Let X be connected, $f \colon X \to \mathbb{R}$ continuous and $a, b \in X$. Then f takes every value between $f(a)$ and $f(b)$.*

Proof. $f(X)$ is connected (Theorem 13.5), hence an interval (Theorem 13.16), and it contains $f(a)$ and $f(b)$. □

13.6 Path-connectedness

Definition 13.18. X is *path-connected* if for all $a, b \in X$ there is a continuous $\gamma\colon [0,1] \to X$ with $\gamma(0) = a$ and $\gamma(1) = b$.

Proposition 13.19. *Every path-connected space is connected.*

Proof. We may assume $X \neq \varnothing$; fix $a \in X$. For each b choose a path γ_b from a to b. Each $\gamma_b([0,1])$ is connected (continuous image of an interval) and contains a, so $X = \bigcup_b \gamma_b([0,1])$ is connected by Theorem 13.6. □

Example 13.20 (Topologist's sine curve)**.** Let $S = \{(x, \sin(1/x)) : x > 0\} \subseteq \mathbb{R}^2$. Then

$$\overline{S} = S \cup (\{0\} \times [-1,1])$$

is connected but not path-connected.

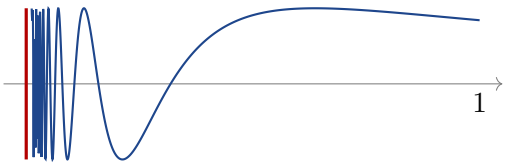


Figure 13.3: The topologist's sine curve. The vertical segment is in the closure but cannot be reached by a path.

Proof. Connected: S is the continuous image of $(0,\infty)$, so it is connected, and so is its closure.

Not path-connected: Suppose $\gamma = (x(t), y(t))\colon [0,1] \to \overline{S}$ is continuous with $\gamma(0) = (1, \sin 1)$ and $\gamma(1) = (0,0)$. Let $a = \inf\{t : x(t) = 0\}$. By continuity $x(a) = 0$, so $a > 0$, and $x(t) > 0$ for $t < a$; hence $y(t) = \sin(1/x(t))$ for $t < a$.

Claim. For every $b \in [-1,1]$ there are $t_n \to a^-$ with $y(t_n) = b$.

Granting the claim, continuity of y gives $y(a) = \lim y(t_n) = b$ for *every* $b \in [-1,1]$, which is absurd.

Proof of claim. Pick $\theta \in [-\pi/2, \pi/2]$ with $\sin\theta = b$, so $\sin(2\pi m + \theta) = b$ for all $m \in \mathbb{Z}$. Given n, let $s_n = a - a/2^n < a$, so $x(s_n) > 0$. Choose $m_n \geq 1$ so large that $0 < \frac{1}{2\pi m_n + \theta} < x(s_n)$. Since $x(a) = 0$, the IVT on $[s_n, a]$ gives $t_n \in (s_n, a)$ with $x(t_n) = \frac{1}{2\pi m_n + \theta}$, whence $y(t_n) = \sin(2\pi m_n + \theta) = b$. As $s_n \to a$, we get $t_n \to a^-$. □

Chapter 14

Compactness

Reading: Munkres, *Topology* (2nd ed.), §§26–29, 37, 38; the Alexander sub-basis theorem is an exercise in §37.

Definition 14.1. Let $E \subseteq X$. A *cover* of E is a family $\mathcal{U} \subseteq 2^X$ with $\bigcup_{A \in \mathcal{U}} A \supseteq E$. It is an *open cover* if every $A \in \mathcal{U}$ is open. A *subcover* is a subfamily $\mathcal{U}' \subseteq \mathcal{U}$ which still covers E.

Definition 14.2. X is *compact* if every open cover of X has a finite subcover. A subset is compact if it is compact as a subspace; equivalently, every cover of E by open sets of X has a finite subcover.

Example 14.3.

- $\mathbb{R}$ is not compact: $\{(-n, n) : n \in \mathbb{N}\}$ has no finite subcover.
- $[0,1]$ is compact (below).
- $\{0,1\}^{\mathbb{N}}$, $[0,1]^{\mathbb{N}}$ and $[0,1]^{\mathbb{R}}$ are compact by Tychonoff's theorem (Theorem 14.13).
- S^1, S^n are compact (closed and bounded in $\mathbb{R}^{n+1}$).
- The Cantor set is compact (closed and bounded in $\mathbb{R}$).

Proposition 14.4. $[0,1]$ *is compact.*

Proof. Let $\mathcal{U}$ be an open cover of $[0,1]$ and

$$S = \{x \in [0,1] : \ [0,x] \text{ is covered by finitely many members of } \mathcal{U}\}.$$

Then $0 \in S$, and $x \in S$, $0 \le y \le x$ imply $y \in S$. Let $e = \sup S$. Choose $A^* \in \mathcal{U}$ with $e \in A^*$ and $\varepsilon > 0$ with $(e-\varepsilon, e+\varepsilon) \cap [0,1] \subseteq A^*$. Some $s \in S$ satisfies $s > e - \varepsilon$; take finitely many $A_1, \dots, A_n \in \mathcal{U}$ covering $[0,s]$. Then $A_1, \dots, A_n, A^*$ cover $[0,e']$ for every $e' \in [0,1]$ with $e' < e + \varepsilon$. So $e \in S$, and every such e' is in S. Since $e = \sup S$, there is no $e' \in (e, e+\varepsilon) \cap [0,1]$; thus $e = 1 \in S$. □

14.1 Basic properties

Proposition 14.5. *If X is compact and $E \subseteq X$ is closed, then E is compact.*

Proof. Let $\mathcal{U}$ be a cover of E by open sets of X. Then $\mathcal{U} \cup \{X \setminus E\}$ is an open cover of X, so there are $A_1, \dots, A_n \in \mathcal{U}$ with $A_1 \cup \cdots \cup A_n \cup (X \setminus E) = X$. Hence $A_1 \cup \cdots \cup A_n \supseteq E$. □

Remark 14.6. A compact subset need not be closed in general: in $\{0,1\}$ with the trivial topology, $\{0\}$ is compact but not closed. In Hausdorff spaces it must be:

Proposition 14.7. *If X is Hausdorff and $E \subseteq X$ is compact, then E is closed.*

Proof. Let $x \in X \setminus E$. For each $y \in E$ choose disjoint open $V_y \ni x$ and $W_y \ni y$. Then $\{W_y\}_{y\in E}$ covers E, so $E \subseteq W_{y_1} \cup \cdots \cup W_{y_n}$ for some $y_1, \dots, y_n$. The set $N = V_{y_1} \cap \cdots \cap V_{y_n}$ is open, contains x, and is disjoint from each W_{y_k}, hence from E. So $X \setminus E$ is open. □

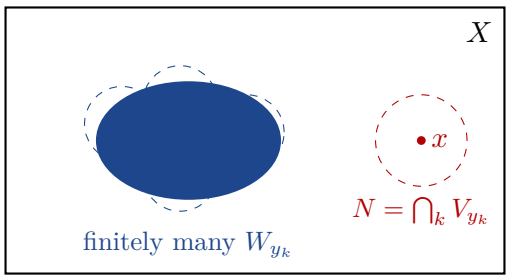


Figure 14.1: Separating a point from a compact set in a Hausdorff space.

Proposition 14.8. *Let X be Hausdorff and $E, F \subseteq X$ disjoint compact sets. Then there are disjoint open sets $N \supseteq E$ and $M \supseteq F$.*

Proof. For each $x \in E$, the proof above (applied to the point x and the compact set F) gives disjoint open $N_x \ni x$ and $M_x \supseteq F$. The N_x cover E; take a finite subcover $N_{x_1}, \dots, N_{x_k}$ and set $N = \bigcup_j N_{x_j}$, $M = \bigcap_j M_{x_j}$. □

Proposition 14.9. *If X is compact and $f\colon X \to Y$ is continuous, then $f(X)$ is compact.*

Proof. If $\mathcal{U}$ is an open cover of $f(X)$, then $\{f^{-1}(B) : B \in \mathcal{U}\}$ is an open cover of X. Choose $B_1, \dots, B_n$ with $\bigcup_k f^{-1}(B_k) = X$; then $B_1, \dots, B_n$ cover $f(X)$. □

Theorem 14.10 (Extreme value theorem)**.** *If X is compact (and non-empty) and $f\colon X \to \mathbb{R}$ is continuous, then f attains a maximum and a minimum.*

Proof. $f(X) \subseteq \mathbb{R}$ is compact, hence closed and bounded (Heine–Borel), so it contains $\sup f(X)$ and $\inf f(X)$. □

Proposition 14.11. *Let $f\colon X \to Y$ be a continuous bijection with X compact and Y Hausdorff. Then f is a homeomorphism.*

Proof. We show f^{-1} is continuous, i.e. $(f^{-1})^{-1}(E) = f(E)$ is closed for closed $E \subseteq X$:

$$E \text{ closed} \overset{14.5}{\Longrightarrow} E \text{ compact} \overset{14.9}{\Longrightarrow} f(E) \text{ compact} \overset{14.7}{\Longrightarrow} f(E) \text{ closed.}$$ □

Example 14.12 (The Cantor set)**.** Let $Y_0 = [0,1]$ and obtain Y_n from Y_{n-1} by deleting the open middle third of each of its intervals. The Cantor set $Y = \bigcap_n Y_n$ is compact (closed in $[0,1]$). The map

$$\phi\colon \{0,1\}^{\mathbb{N}} \to Y, \qquad \phi(x_1, x_2, \dots) = \sum_{n=1}^{\infty} \frac{2x_n}{3^n}$$

is a continuous bijection from a compact space to a Hausdorff space, hence a homeomorphism. So the Cantor set "is" Cantor space.

14.2 Tychonoff's theorem

Theorem 14.13 (Tychonoff)**.** *A product of compact spaces is compact. (Conversely, if $\prod_i X_i$ is compact and non-empty, each $X_i = \pi_i(\prod X_j)$ is compact.)*

Example 14.14. Closed cubes $[0,1]^n$; Cantor space $\{0,1\}^{\mathbb{N}}$; the torus $S^1 \times S^1$; the huge cube $[0,1]^{\mathbb{R}}$.

Proof for two factors (tube lemma). Let X, Y be compact and $\mathcal{C}$ an open cover of $X \times Y$. Fix $y \in Y$. The slice $X \times \{y\} \cong X$ is compact, so finitely many members $\mathcal{C}_y \subseteq \mathcal{C}$ cover it; let $V_y = \bigcup_{A \in \mathcal{C}_y} A$, an open set containing $X \times \{y\}$.

Claim (Tube lemma). There is an open $B_y \ni y$ with $X \times B_y \subseteq V_y$.

Proof of claim. For each $x \in X$ choose open $I_x \subseteq X$ and $J_x \subseteq Y$ with $(x,y) \in I_x \times J_x \subseteq V_y$. The sets I_x cover the compact X, so $X = I_{x_1} \cup \cdots \cup I_{x_k}$. Let $B_y = J_{x_1} \cap \cdots \cap J_{x_k}$, open and containing y. For $(x', y') \in X \times B_y$ pick j with $x' \in I_{x_j}$; then $(x', y') \in I_{x_j} \times J_{x_j} \subseteq V_y$.

Now $\{B_y\}_{y \in Y}$ is an open cover of Y; choose $y_1, \ldots, y_m$ with $Y = \bigcup_l B_{y_l}$. Then

$$X \times Y = \bigcup_l X \times B_{y_l} \subseteq \bigcup_l V_{y_l} = \bigcup_l \bigcup_{A \in \mathcal{C}_{y_l}} A,$$

so the finite family $\bigcup_l \mathcal{C}_{y_l}$ covers $X \times Y$. □

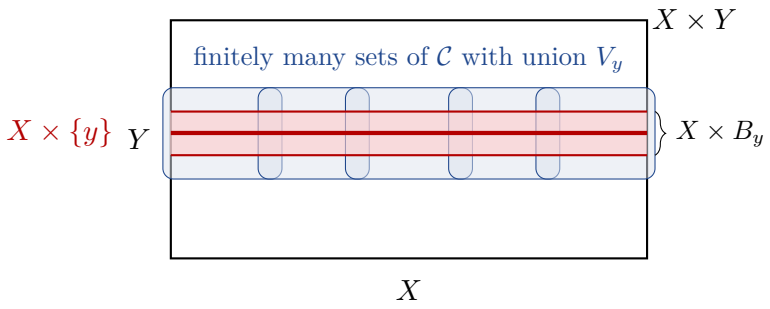


Figure 14.2: The tube lemma: a tube $X \times B_y$ fits inside V_y.

14.3 Compactness via bases and sub-bases

Proposition 14.15. *Let $\mathcal{B}$ be a basis for X and $E \subseteq X$. Then E is compact if and only if every cover of E by members of $\mathcal{B}$ has a finite subcover.*

Proof. ($\Rightarrow$) Basic sets are open. ($\Leftarrow$) Let $\mathcal{U} = \{A_i\}_{i \in I}$ be an open cover of E. For each i and $x \in A_i$ choose $B_{i,x} \in \mathcal{B}$ with $x \in B_{i,x} \subseteq A_i$. Then $\{B_{i,x}\}$ is a $\mathcal{B}$-cover of E; take a finite subcover and, for each $B_{i,x}$ in it, the corresponding A_i. These finitely many A_i cover E. □

Theorem 14.16 (Alexander sub-basis theorem)**.** *Let $\mathcal{S}$ be a sub-basis for X. Then X is compact if and only if every cover of X by members of $\mathcal{S}$ has a finite subcover.*

Example 14.17 ($[0,1]$ via rays)**.** The open rays $\mathcal{S} = \{(-\infty, b)\} \cup \{(a, \infty)\}$ form a sub-basis of $\mathbb{R}$, so their traces on $[0,1]$ form a sub-basis of $[0,1]$. Let $\mathcal{C}$ be a cover of $[0,1]$ by rays, and put $\beta = \sup\{b : (-\infty, b) \in \mathcal{C}\}$ and $\alpha = \inf\{a : (a, \infty) \in \mathcal{C}\}$ (with $\sup \varnothing = -\infty$, $\inf \varnothing = +\infty$).

- If $\alpha < \beta$, some $(a, \infty), (-\infty, b) \in \mathcal{C}$ have $a < b$; these two cover $\mathbb{R}$.
- If $\beta > 1$, a single left ray covers $[0,1]$; if $\alpha < 0$, a single right ray does.
- Otherwise $\beta \le \alpha$, $\beta \le 1$ and $\alpha \ge 0$. Then $t = \max\{\beta, 0\} \in [0,1]$ satisfies $t \ge b$ for every left ray and $t \le a$ for every right ray, so t is not covered: impossible.

So $\mathcal{C}$ has a finite subcover, and $[0,1]$ is compact by Theorem 14.16.

Proof of Tychonoff's theorem from Alexander's theorem. Let $X = \prod_i X_i$ with all X_i compact, and let $\mathcal{C}$ be a cover of X by sub-basic sets $\pi_i^{-1}(U)$ (Theorem 5.4). For each i let

$$\mathcal{U}_i = \{U \subseteq X_i \text{ open} : \ \pi_i^{-1}(U) \in \mathcal{C}\}.$$

Case 1: some $\mathcal{U}_i$ covers X_i. By compactness of X_i a finite $\mathcal{U}_i^0 \subseteq \mathcal{U}_i$ covers X_i, and then $\{\pi_i^{-1}(U) : U \in \mathcal{U}_i^0\} \subseteq \mathcal{C}$ is a finite subcover of X.

Case 2: no $\mathcal{U}_i$ covers X_i. Choose (axiom of choice) $x_i \in X_i$ not covered by $\mathcal{U}_i$ for each i, and let $x = (x_i)_{i \in I}$. If $x \in \pi_i^{-1}(U) \in \mathcal{C}$, then $x_i \in U \in \mathcal{U}_i$, contradicting the choice of x_i. So x is not covered by $\mathcal{C}$, which is impossible.

Hence Case 1 occurs, and X is compact by Theorem 14.16. □

14.4 Compactness and sequences

Definition 14.18. X is *sequentially compact* if every sequence in X has a subsequence converging to a point of X.

Remark 14.19. For $E \subseteq \mathbb{R}^n$ the following are equivalent: (1) E is compact; (2) E is sequentially compact; (3) E is closed and bounded (Heine–Borel). For general spaces (1) and (2) are *not* equivalent.

Example 14.20. (1) $\{0,1\}^{\mathbb{R}}$ is compact (Tychonoff) but not sequentially compact. Identify $\mathbb{R}$ with a set of binary sequences (they have the same cardinality) and let $x_t^{(n)}$ be the n-th digit of t. Given any subsequence $n_1 < n_2 < \cdots$, choose t whose digits at positions $n_1, n_2, n_3, \ldots$ are $0, 1, 0, 1, \ldots$; then $x_t^{(n_k)}$ does not converge, so $x^{(n_k)}$ does not converge (convergence is coordinatewise).

(2) $\Sigma = \{x \in \{0,1\}^{\mathbb{R}} : \ x_t = 0 \text{ for all but countably many } t\}$ is sequentially compact but not compact. Sequential compactness: a sequence in Σ has only countably many coordinates that are ever non-zero, so it lives in a copy of $\{0,1\}^{\mathbb{N}}$, which is sequentially compact (Theorem 14.49). Not compact: Σ is dense in the Hausdorff space $\{0,1\}^{\mathbb{R}}$ but not equal to it, hence not closed, hence not compact (Theorem 14.7).

The space Σ also shows that closures are not determined by sequences in general: the all-ones point is in $\mathrm{cl}(\Sigma)$, but no sequence in Σ converges to it (a limit of a sequence in Σ is in Σ).

Theorem 14.21. *A metrizable space X is compact if and only if it is sequentially compact.*

Proof. Fix a compatible metric d.

($\Rightarrow$) Suppose (x_n) has no convergent subsequence.

Claim 1. Every $x \in X$ has an $\varepsilon_x > 0$ such that $x_n \in B_{\varepsilon_x}(x)$ for only finitely many n.

Proof. Otherwise some x has $x_n \in B_{1/k}(x)$ for infinitely many n, for every k. Choose $n_1 < n_2 < \cdots$ inductively with $x_{n_k} \in B_{1/k}(x)$. Then $x_{n_k} \to x$, a contradiction.

The balls $\{B_{\varepsilon_x}(x)\}_{x \in X}$ cover X. A finite subfamily would contain x_n for only finitely many indices n in total, so it cannot cover the whole sequence. Thus X is not compact.

($\Leftarrow$) Let X be sequentially compact and $\mathcal{U}$ an open cover.

Claim 2 (Lebesgue number). There is $\varepsilon > 0$ such that every ball $B_\varepsilon(x)$ lies inside some member of $\mathcal{U}$.

Proof. If not, for each n pick x_n such that $B_{1/n}(x_n)$ lies in no member of $\mathcal{U}$. A subsequence $x_{n_k} \to x$. Choose $A^* \in \mathcal{U}$ and $\varepsilon > 0$ with $B_\varepsilon(x) \subseteq A^*$, then k with $1/n_k < \varepsilon/2$ and $d(x_{n_k}, x) < \varepsilon/2$. For $y \in B_{1/n_k}(x_{n_k})$, $d(y, x) < \varepsilon/2 + \varepsilon/2 = \varepsilon$, so $B_{1/n_k}(x_{n_k}) \subseteq A^*$. Contradiction.

Claim 3 (Total boundedness). For every $\varepsilon > 0$, X is covered by finitely many ε-balls.

Proof. If not, pick x_1, then $x_2 \notin B_\varepsilon(x_1)$, then $x_3 \notin B_\varepsilon(x_1) \cup B_\varepsilon(x_2)$, and so on. Then $d(x_i, x_j) \geq \varepsilon$ for $i \neq j$, so no subsequence is Cauchy, hence none converges. Contradiction.

With ε from Claim 2, cover X by $B_\varepsilon(x_1), \ldots, B_\varepsilon(x_n)$ and choose $A_i \in \mathcal{U}$ with $B_\varepsilon(x_i) \subseteq A_i$. Then $A_1, \ldots, A_n$ is a finite subcover. □

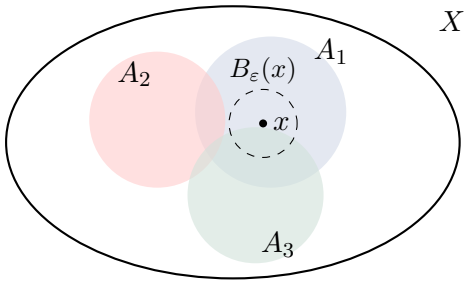


Figure 14.3: A Lebesgue number: every ε-ball fits in one member of the cover.

Definition 14.22. A metric space is *totally bounded* if for every $\varepsilon > 0$ it is covered by finitely many ε-balls.

Theorem 14.23. *A metric space E is compact if and only if it is complete and totally bounded.*

Proof. ($\Rightarrow$) Totally bounded: the cover by all ε-balls has a finite subcover. Complete: a Cauchy sequence has a convergent subsequence (Theorem 14.21), and a Cauchy sequence with a convergent subsequence converges.

($\Leftarrow$) By Theorem 14.21 it suffices to show sequential compactness. Let (x_n) be a sequence. Cover E by finitely many balls of radius 1; one of them contains x_n for infinitely many n, say for $n \in I_1$. Cover by finitely many balls of radius $1/2$; one contains x_n for infinitely many $n \in I_1$, say $n \in I_2 \subseteq I_1$. Continue to get infinite sets $I_1 \supseteq I_2 \supseteq \cdots$ such that $\{x_n : n \in I_k\}$ lies in a ball of radius $1/k$. Choose $n_1 < n_2 < \cdots$ with $n_k \in I_k$. For $j, l \geq k$, both x_{n_j}, x_{n_l} lie in the same $1/k$-ball, so $d(x_{n_j}, x_{n_l}) < 2/k$. The subsequence is Cauchy and converges by completeness. □

Theorem 14.24 (Lebesgue number lemma)**.** *Let X be a compact metric space and $\mathcal{U}$ an open cover. There is $\delta > 0$ such that every $C \subseteq X$ with $\operatorname{diam}(C) < \delta$ lies inside some member of $\mathcal{U}$. (Here $\operatorname{diam}(C) = \sup\{d(x,y) : x, y \in C\}$.)*

Proof. X is sequentially compact (Theorem 14.21), so Claim 2 gives $\varepsilon > 0$ such that every ε-ball lies in a member of $\mathcal{U}$. Take $\delta = \varepsilon$. If $C \neq \varnothing$ has $\operatorname{diam}(C) < \varepsilon$, pick $x \in C$; then $C \subseteq B_\varepsilon(x)$. □

Proposition 14.25. *If X, Y are metric spaces, X is compact and $f\colon X \to Y$ is continuous, then f is uniformly continuous.*

Proof. Let $\varepsilon > 0$. For each $z \in X$ choose $\delta_z > 0$ with $f(B_{\delta_z}(z)) \subseteq B_{\varepsilon/2}(f(z))$. Let δ be a Lebesgue number for the cover $\{B_{\delta_z}(z)\}_{z \in X}$. If $d(x,y) < \delta$, then $\operatorname{diam}\{x,y\} < \delta$, so $x, y \in B_{\delta_z}(z)$ for some z, and

$$d(f(x), f(y)) \le d(f(x), f(z)) + d(f(z), f(y)) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$ □

14.5 Compactness via closed sets

Definition 14.26. A family $\mathcal{C} \subseteq 2^X$ has the *finite intersection property* (FIP) if every finite subfamily has non-empty intersection: $C_1, \dots, C_n \in \mathcal{C} \implies C_1 \cap \cdots \cap C_n \neq \varnothing$.

Proposition 14.27. *X is compact if and only if every family of closed sets with the FIP has non-empty intersection.*

Proof. Pass to complements: $\{C_i\}$ closed with $\bigcap_i C_i = \varnothing \iff \{X \setminus C_i\}$ is an open cover. "Every open cover has a finite subcover" then reads "every family of closed sets with empty intersection has a finite subfamily with empty intersection", which is the contrapositive of the statement. □

Example 14.28 (Compactness in logic)**.** Let V be a set of Boolean variables and Σ a set of propositional formulas over V. If every finite subset of Σ is satisfiable, then Σ is satisfiable.

Proof sketch. Truth assignments form the compact space $\{0,1\}^V$ (Tychonoff). For a formula φ the set M_φ of assignments satisfying φ depends on finitely many coordinates, so it is clopen. By hypothesis $\{M_\varphi : \varphi \in \Sigma\}$ has the FIP, so by Theorem 14.27 some assignment lies in every M_φ.

Corollary 14.29 (Nested closed sets)**.** *Let X be compact and $C_1 \supseteq C_2 \supseteq \cdots$ non-empty closed sets. Then $\bigcap_n C_n \neq \varnothing$.*

Proof. A finite subfamily has intersection $C_{\max} \neq \varnothing$, so $\{C_n\}$ has the FIP. □

Example 14.30. Compactness is needed:

- in $X = \mathbb{R}$, $C_n = [n, \infty)$ has $\bigcap_n C_n = \varnothing$;
- in $X = \mathbb{Q}$, $C_n = [\sqrt{2} - \tfrac{1}{n}, \sqrt{2} + \tfrac{1}{n}] \cap \mathbb{Q}$ has $\bigcap_n C_n = \varnothing$.

14.6 Local compactness

Definition 14.31. A *compact neighbourhood* of x is a compact set C with $x \in \operatorname{int}(C)$. X is *locally compact* if every point has a compact neighbourhood.

Lemma 14.32. *Let X be Hausdorff and $A, B \subseteq X$ disjoint compact sets. Then there is an open $N \supseteq A$ with $\operatorname{cl}(N) \cap B = \varnothing$.*

Proof. By Theorem 14.8 take disjoint open $N \supseteq A$ and $M \supseteq B$. Then $X \setminus M$ is a closed set containing N, so $\operatorname{cl}(N) \subseteq X \setminus M$, which misses B. □

Proposition 14.33. *Let X be Hausdorff. Then X is locally compact if and only if every point has a neighbourhood basis of compact neighbourhoods, i.e. for every open $A \ni x$ there is a compact K with $x \in \operatorname{int}(K) \subseteq K \subseteq A$.*

Proof. ($\Leftarrow$) Take $A = X$.

($\Rightarrow$) Let C be compact with $x \in \operatorname{int}(C)$, and let $A \ni x$ be open. Put $W = A \cap \operatorname{int}(C)$, open with $x \in W$. The set $C \setminus W$ is closed in C, hence compact, and does not contain x. By Theorem 14.32 (with $A = \{x\}$) there is an open $N \ni x$ with $\operatorname{cl}(N) \cap (C \setminus W) = \varnothing$. Let $D = N \cap W$ and $K = \operatorname{cl}(D)$. Then:

- $x \in D \subseteq \operatorname{int}(K)$;
- $K \subseteq \operatorname{cl}(N) \cap C$ (since C is closed, by Theorem 14.7), and $\operatorname{cl}(N) \cap C \subseteq W \subseteq A$;
- K is closed and contained in C, hence compact. □

Proposition 14.34. *Let X be locally compact Hausdorff.*

(1) *If $A \subseteq X$ is closed, then A is locally compact.*

(2) *If $U \subseteq X$ is open, then U is locally compact.*

(3) *If $A, B \subseteq X$ are locally compact (as subspaces), then so is $A \cap B$.*

Proof. (1) Let $x \in A$ and C a compact neighbourhood of x in X. Then $C \cap A$ is compact (closed in C), and it contains $\operatorname{int}(C) \cap A$, which is open in A and contains x.

(2) By Theorem 14.33 there is a compact K with $x \in \operatorname{int}(K) \subseteq K \subseteq U$; this is a compact neighbourhood of x in U.

(3) Let $x \in A \cap B$, with compact neighbourhoods C_A of x in A and C_B of x in B; so $C_A \supseteq O_A \cap A$ and $C_B \supseteq O_B \cap B$ with O_A, O_B open in X containing x. C_A is compact in a Hausdorff space, hence closed, so $C_A \cap C_B$ is closed in C_B, hence compact. It contains $(O_A \cap O_B) \cap (A \cap B)$, which is open in $A \cap B$ and contains x. □

Remark 14.35. A continuous image of a locally compact space need not be locally compact. For example, the identity map from $\mathbb{Q}$ with the discrete topology (locally compact: $\{q\}$ is a compact neighbourhood of q) to $\mathbb{Q}$ with the usual topology is continuous and onto, but $\mathbb{Q}$ is not locally compact.

Definition 14.36. $f\colon X \to Y$ is an *open map* if $f(U)$ is open for every open $U \subseteq X$.

Proposition 14.37. *If X is locally compact and $f\colon X \to Y$ is continuous and open, then $f(X)$ is locally compact.*

Proof. Let $y = f(x)$ and let C be a compact neighbourhood of x. Then $f(C)$ is compact, $f(\operatorname{int}(C))$ is open, and $y \in f(\operatorname{int}(C)) \subseteq f(C)$. So $f(C)$ is a compact neighbourhood of y. □

14.7 Compactifications

Example 14.38.

- $\mathbb{R}$ is not compact, but $\mathbb{R} \cup \{\infty\} \cong S^1$: stereographic projection identifies $\mathbb{R}$ with $S^1 \setminus \{p\}$, and ∞ fills in p.
- Likewise $\mathbb{R}^n \cong S^n \setminus \{p\}$, so $\mathbb{R}^n \cup \{\infty\} \cong S^n$.
- $\mathbb{R} \cup \{-\infty, +\infty\} \cong [0,1]$ (the two-point compactification), e.g. via $x \mapsto \frac{1}{2} + \frac{1}{\pi}\arctan x$.

Definition 14.39. A *compactification* of a space X is a pair $(\tilde{X}, \phi)$ where $\tilde{X}$ is compact Hausdorff and $\phi\colon X \to \tilde{X}$ is an embedding with $\phi(X)$ dense in $\tilde{X}$.

The one-point compactification

Let X be locally compact, Hausdorff and not compact. Put $X^* = X \cup \{\infty\}$, $\phi^*(x) = x$, and

$$\mathcal{T}^* = \mathcal{T} \ \cup \ \{X^* \setminus C : \ C \subseteq X \text{ compact}\}.$$

Exercise 14.40. Check that $\mathcal{T}^*$ is a topology on X^* and that ϕ^* is an embedding.

Remark 14.41. $\phi^*(X)$ is dense: an open set containing ∞ has the form $X^* \setminus C$ with C compact, and since X is not compact, $X \setminus C \neq \varnothing$.

Proposition 14.42. *X^* is compact.*

Proof. Let $\mathcal{U}$ be an open cover of X^* and pick $U \in \mathcal{U}$ with $\infty \in U$; then $U = X^* \setminus C$ with $C \subseteq X$ compact. Finitely many members of $\mathcal{U}$ cover C; together with U they cover X^*. □

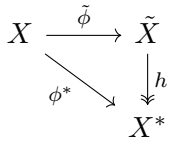
Proposition 14.43. *X^* is Hausdorff.*

Proof. Let $x \neq y$ in X^*. If $x, y \in X$, separate them by disjoint open sets of X, which are open in X^*. If $y = \infty$, let C be a compact neighbourhood of x in X; then $\operatorname{int}(C) \ni x$ and $X^* \setminus C \ni \infty$ are disjoint and open. □

Theorem 14.44 (Universality of the one-point compactification)**.** *Let X be non-compact, locally compact and Hausdorff, and let $(\tilde{X}, \tilde{\phi})$ be any compactification of X.*

(1) *There is a quotient map $h\colon \tilde{X} \to X^*$ with $h \circ \tilde{\phi} = \phi^*$.*

(2) *If $\tilde{X} \setminus \tilde{\phi}(X)$ is a single point, h is a homeomorphism. In particular the one-point compactification is unique up to homeomorphism.*



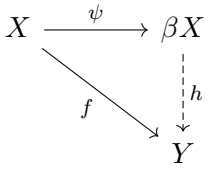
Sketch. Define $h(y) = \tilde{\phi}^{-1}(y)$ for $y \in \tilde{\phi}(X)$ and $h(y) = \infty$ otherwise; it is onto and $h \circ \tilde{\phi} = \phi^*$. For continuity: if $C \subseteq X$ is compact, $h^{-1}(X^* \setminus C) = \tilde{X} \setminus \tilde{\phi}(C)$ is open since $\tilde{\phi}(C)$ is compact, hence closed. If $U \subseteq X$ is open, $h^{-1}(U) = \tilde{\phi}(U)$, which is open in $\tilde{\phi}(X)$, and $\tilde{\phi}(X)$ is open in $\tilde{X}$ (a locally compact dense subspace of a Hausdorff space is open). A continuous surjection from a compact space onto a Hausdorff space maps closed sets to closed sets, so it is a quotient map. If h is also injective, it is a homeomorphism by Theorem 14.11. □

The Stone–Čech compactification

Definition 14.45. X is *completely regular* if singletons are closed and for every point p and closed $C \not\ni p$ there is a continuous $f\colon X \to [0,1]$ with $f(p) = 0$ and $f(C) = \{1\}$.

Theorem 14.46. *X embeds into a cube $[0,1]^J$ if and only if X is completely regular.*

Remark 14.47. If $\psi\colon X \to [0,1]^J$ is the embedding obtained from *all* of $C(X,[0,1])$ (Theorem 17.3), then $\beta X = \operatorname{cl}(\psi(X)) \subseteq [0,1]^J$ is compact (closed in a compact cube) and $(\beta X, \psi)$ is the *Stone–Čech compactification* of X. It is characterised by a universal property: every continuous $f\colon X \to Y$ into a compact Hausdorff space Y extends uniquely to a continuous $h\colon \beta X \to Y$ with $h \circ \psi = f$.



Example 14.48. $\mathbb{R} \cup \{\pm\infty\} \cong [0,1]$ and $\mathbb{N} \cup \{\infty\} \cong \{1/n : n \in \mathbb{N}\} \cup \{0\}$ are compactifications of $\mathbb{R}$ and $\mathbb{N}$, but they are *not* the Stone–Čech compactifications. For instance $\sin x$ does not extend to $[-\infty,\infty]$, and $n \mapsto (-1)^n$ does not extend to $\mathbb{N} \cup \{\infty\}$. In fact $\beta\mathbb{N}$ has cardinality $2^{\mathfrak{c}}$.

14.8 Countable products of sequentially compact spaces

Theorem 14.49. *A countable product of sequentially compact spaces is sequentially compact.*

Proof. Let $x^{(1)}, x^{(2)}, \dots$ be a sequence in $X = \prod_{k=1}^{\infty} X_k$, with $x^{(n)} = (x_1^{(n)}, x_2^{(n)}, \dots)$. Since X_1 is sequentially compact there is an *infinite* $I_1 \subseteq \mathbb{N}$ with $x_1^{(n)} \to y_1$ as $n \to \infty$, $n \in I_1$. Since X_2 is sequentially compact there is an infinite $I_2 \subseteq I_1$ with $x_2^{(n)} \to y_2$ along I_2. Continuing, we get infinite sets $I_1 \supseteq I_2 \supseteq \cdots$ with $x_k^{(n)} \to y_k$ along I_k.

Diagonal step. Let n_k be the least element of I_k larger than n_{k-1}, and $I = \{n_1 < n_2 < \cdots\}$. For each k, $I \setminus I_k \subseteq \{n_1, \dots, n_{k-1}\}$ is finite, so $x_k^{(n)} \to y_k$ along I for every k. Since convergence in the product is coordinatewise, $x^{(n)} \to y = (y_k)_k$ along I. □

Chapter 15

Separation and Countability Axioms

Reading: Munkres, *Topology* (2nd ed.), §§30–32.

15.1 Separation axioms

Definition 15.1. A space X is

- T_1 if singletons are closed;
- T_2 **(Hausdorff)** if distinct points have disjoint open neighbourhoods;
- T_3 **(regular)** if it is T_1 and for every $x \in X$ and closed $C \not\ni x$ there are disjoint open sets $A \ni x$ and $B \supseteq C$;
- $T_{3\frac{1}{2}}$ **(completely regular)** if it is T_1 and for every $x \in X$ and closed $C \not\ni x$ there is a continuous $f\colon X \to [0,1]$ with $f(x) = 0$ and $f(C) = \{1\}$;
- T_4 **(normal)** if it is T_1 and for all disjoint closed C, D there are disjoint open sets $A \supseteq C$ and $B \supseteq D$.

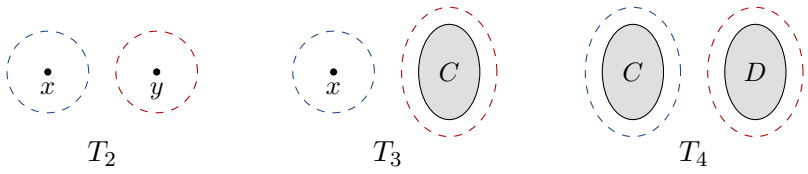


Figure 15.1: What the separation axioms separate: points, a point and a closed set, two closed sets.

Remark 15.2. $T_4 \implies T_{3\frac{1}{2}} \implies T_3 \implies T_2 \implies T_1$. The first implication is Urysohn's lemma (Theorem 16.1); for the second, take $A = f^{-1}([0, \frac{1}{2}))$ and $B = f^{-1}((\frac{1}{2}, 1])$; the third uses that singletons are closed.

Example 15.3.

- $\mathbb{R}$, and more generally every metrizable space, is normal (Theorem 15.7).
- Every compact Hausdorff space is normal.
- $\mathbb{R}^{\mathbb{R}} \cong (0,1)^{\mathbb{R}}$ is regular (a product of regular spaces) but not normal.

15.2 Countability axioms

Definition 15.4. A space X is

- *first countable* if every point has a countable neighbourhood basis (Theorem 9.6);
- *second countable* if the topology has a countable basis;

- *separable* if it has a countable dense subset.

Example 15.5.

- $\mathbb{R}$ is second countable (intervals with rational endpoints) and separable ($\mathbb{Q}$).
- Every metrizable space is first countable.
- $\mathbb{R}/\mathbb{Z}$ — meaning $\mathbb{R}$ with all integers collapsed to a single point (not the circle group!) — is not first countable at the collapsed point.
- $\{0,1\}^{\mathbb{R}}$ is not first countable (Theorem 9.8).
- $\mathbb{R}$ with the discrete topology is first countable but not second countable (each singleton must be a basis element) and not separable.
- If X is a compact metric space, $C(X,\mathbb{R})$ with the sup norm is a separable Banach space.

Remark 15.6. A *Banach space* is a normed vector space $(V, \|\cdot\|)$ over $\mathbb{R}$ or $\mathbb{C}$ that is complete for the metric $d(u,v) = \|u - v\|$.

15.3 Which properties are inherited?

	Subspace	**Product**	**Quotient**
Hausdorff	✓	✓	✗
Regular	✓	✓	✗
Normal	✗ (✓ for closed subspaces)	✗	✗
First countable	✓	✓ countable / ✗ uncountable	✗
Second countable	✓	✓ countable / ✗ uncountable	✗
Separable	✗ (✓ for open subspaces)	✓ up to $\mathfrak{c}$ factors	✓

Table 15.1: Inheritance of separation and countability properties. "Uncountable" means infinitely many non-trivial factors, uncountably many. Separability passes to continuous images, in particular to quotients.

15.4 Some implications

Proposition 15.7. (1) *Metrizable* $\Longrightarrow$ *first countable.*

(2) *Metrizable and separable* $\Longrightarrow$ *second countable.*

(3) *Second countable* $\Longrightarrow$ *separable.*

(4) *Compact Hausdorff* $\Longrightarrow$ *normal.*

(5) *Metrizable* $\Longrightarrow$ *normal.*

(6) *Regular and second countable* $\Longrightarrow$ *normal.*

Proofs of (1)–(4). (1) $\{B_{1/n}(x)\}_n$. (2) If D is countable and dense, $\mathcal{B} = \{B_q(x) : x \in D,\ q \in \mathbb{Q}^+\}$ is a countable basis: given $y \in U$ open, take $B_{2q}(y) \subseteq U$, then $x \in D \cap B_q(y)$, and $y \in B_q(x) \subseteq B_{2q}(y) \subseteq U$. (3) Pick one point from each non-empty member of a countable basis. (4) Closed subsets of a compact space are compact, and disjoint compact sets in a Hausdorff space are separated by Theorem 14.8. □

Proof of (5). Fix a compatible metric d and disjoint closed P, Q (both non-empty; otherwise take X and $\varnothing$). For $x \in P$ choose $\varepsilon_x > 0$ with $B_{\varepsilon_x}(x) \cap Q = \varnothing$, and for $y \in Q$ choose $\delta_y > 0$ with $B_{\delta_y}(y) \cap P = \varnothing$. Let

$$U = \bigcup_{x \in P} B_{\varepsilon_x/2}(x), \qquad W = \bigcup_{y \in Q} B_{\delta_y/2}(y).$$

These are open and contain P and Q. If $z \in B_{\varepsilon_x/2}(x) \cap B_{\delta_y/2}(y)$, then $d(x,y) < \frac{\varepsilon_x + \delta_y}{2} \le \max\{\varepsilon_x, \delta_y\}$. If $\varepsilon_x \ge \delta_y$ this gives $y \in B_{\varepsilon_x}(x)$, contradicting $B_{\varepsilon_x}(x) \cap Q = \varnothing$; otherwise $x \in B_{\delta_y}(y)$, a contradiction too. So $U \cap W = \varnothing$. □

Proof of (6). Let A, B be disjoint closed sets, both non-empty, and $\mathcal{B}$ a countable basis. For $x \in A$, regularity gives open $U_x \ni x$ and $W_x \supseteq B$ with $U_x \cap W_x = \varnothing$; choose $F \in \mathcal{B}$ with $x \in F \subseteq U_x$. Then $\operatorname{cl}(F) \subseteq X \setminus W_x$, so $\operatorname{cl}(F) \cap B = \varnothing$. Only countably many basis elements arise this way, so:

- there are $F_1, F_2, \cdots \in \mathcal{B}$ with $\bigcup_n F_n \supseteq A$ and $\operatorname{cl}(F_n) \cap B = \varnothing$ for all n;
- similarly there are $G_1, G_2, \cdots \in \mathcal{B}$ with $\bigcup_n G_n \supseteq B$ and $\operatorname{cl}(G_n) \cap A = \varnothing$ for all n.

Define

$$F_n' = F_n \setminus \bigcup_{i=1}^{n} \operatorname{cl}(G_i), \qquad G_n' = G_n \setminus \bigcup_{i=1}^{n} \operatorname{cl}(F_i),$$

which are open (open minus closed), and set $\tilde{A} = \bigcup_n F_n'$, $\tilde{B} = \bigcup_n G_n'$.

- $A \subseteq \tilde{A}$: if $a \in A$, then $a \in F_n$ for some n, and $a \notin \operatorname{cl}(G_i)$ for all i, so $a \in F_n'$. Similarly $B \subseteq \tilde{B}$.
- $\tilde{A} \cap \tilde{B} = \varnothing$: suppose $z \in F_n' \cap G_m'$. If $n \le m$, then $z \in G_m'$ gives $z \notin \operatorname{cl}(F_n) \supseteq F_n$, but $z \in F_n' \subseteq F_n$. If $m \le n$, argue symmetrically. □

Chapter 16

Urysohn's Lemma and Tietze's Extension Theorem

Reading: Munkres, *Topology* (2nd ed.), §§33, 35.

16.1 Urysohn's lemma

Theorem 16.1 (Urysohn's lemma)**.** *Let X be normal and $A, B \subseteq X$ disjoint closed sets. Then there is a continuous $f\colon X \to [0,1]$ with $f(A) \subseteq \{0\}$ and $f(B) \subseteq \{1\}$.*

Proof when X is metrizable. Let d be a compatible metric; we may assume $A, B \neq \varnothing$. Put

$$f(x) = \frac{d(x,A)}{d(x,A) + d(x,B)}, \qquad d(x,A) = \inf_{a \in A} d(x,a).$$

The map $x \mapsto d(x,A)$ is continuous (it is 1-Lipschitz), and $d(x,A) = 0$ iff $x \in \mathrm{cl}(A) = A$. Since $A \cap B = \varnothing$, the denominator never vanishes, so f is continuous; clearly $f = 0$ on A and $f = 1$ on B. □

The general proof builds the level sets of f by hand. We need a reformulation of normality.

Lemma 16.2. *A T_1 space X is normal if and only if for every closed A and open $W \supseteq A$ there is an open U with*

$$A \subseteq U \subseteq \mathrm{cl}(U) \subseteq W.$$

Proof. ($\Rightarrow$) Separate A and $X \setminus W$ by disjoint open $U \supseteq A$, $V \supseteq X \setminus W$. Then $U \subseteq X \setminus V$, a closed set, so $\mathrm{cl}(U) \subseteq X \setminus V \subseteq W$. ($\Leftarrow$) Given disjoint closed A, B, apply this with $W = X \setminus B$; then U and $X \setminus \mathrm{cl}(U)$ separate A and B. □

Proof of Urysohn's lemma for normal X. Let $P = \{k/2^n : n \geq 0,\ 0 \leq k \leq 2^n\}$ be the dyadic rationals in $[0,1]$; they are dense in $[0,1]$.

Step 1: open sets indexed by P. We construct open sets U_r ($r \in P$) such that

$$A \subseteq U_0, \qquad U_1 = X \setminus B, \qquad r < s \implies \mathrm{cl}(U_r) \subseteq U_s. \tag{16.1}$$

Let $U_1 = X \setminus B$, open and containing A. By Theorem 16.2 choose U_0 open with $A \subseteq U_0 \subseteq \mathrm{cl}(U_0) \subseteq U_1$. Suppose the U_r are defined for all $r = k/2^{n-1}$. For odd k, apply Theorem 16.2 to the closed set $\mathrm{cl}(U_{(k-1)/2^n})$ inside the open set $U_{(k+1)/2^n}$ (these indices have denominator 2^{n-1}) to get $U_{k/2^n}$ with

$$\mathrm{cl}(U_{(k-1)/2^n}) \subseteq U_{k/2^n} \subseteq \mathrm{cl}(U_{k/2^n}) \subseteq U_{(k+1)/2^n}.$$

By induction, (16.1) holds for all $r < s$ in P.

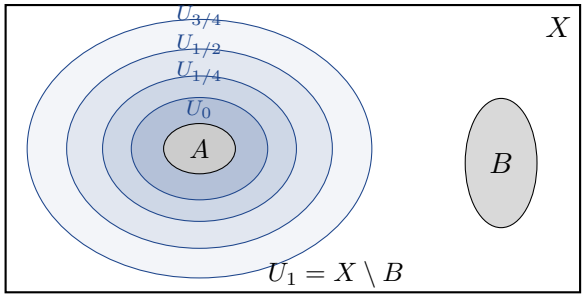


Figure 16.1: Nested open sets U_r between A and $X \setminus B$; $f(x)$ is the "first level" at which x enters.

Step 2: the function. Define

$$f(x) = \inf\{r \in P : \ x \in U_r\}, \qquad \text{with } \inf \varnothing = 1.$$

If $x \in A$ then $x \in U_0$, so $f(x) = 0$. If $x \in B$ then $x \notin U_1 \supseteq U_r$ for all r, so $f(x) = 1$.

Step 3: two observations. For $r \in P$:

(i) if $x \in \mathrm{cl}(U_r)$ then $f(x) \le r$: indeed $x \in U_s$ for every $s \in P$ with $s > r$, and such s come arbitrarily close to r;

(ii) if $x \notin U_r$ then $f(x) \ge r$: indeed $x \notin U_s$ for every $s \le r$.

Step 4: continuity. It suffices to show that for every x and every open interval $(c, d) \ni f(x)$ there is an open $W \ni x$ with $f(W) \subseteq (c, d)$ (intersected with $[0, 1]$).

- If $0 < f(x) < 1$: pick $r, s \in P$ with $c < r < f(x) < s < d$. Let $W = U_s \setminus \mathrm{cl}(U_r)$, which is open. Since $f(x) < s$, there is $s' < s$ in P with $x \in U_{s'} \subseteq U_s$; and $x \notin \mathrm{cl}(U_r)$ by (i) since $f(x) > r$. For $y \in W$, (i) gives $f(y) \le s$ and (ii) gives $f(y) \ge r$, so $f(y) \in [r, s] \subseteq (c, d)$.
- If $f(x) = 0$: pick $s \in P$ with $0 < s < d$, and let $W = U_s \ni x$. For $y \in W$, $f(y) \le s < d$.
- If $f(x) = 1$: pick $r \in P$ with $c < r < 1$, and let $W = X \setminus \mathrm{cl}(U_r) \ni x$ (by (i)). For $y \in W$, $f(y) \ge r > c$ by (ii).

Hence f is continuous. □

16.2 Tietze's extension theorem

Theorem 16.3 (Tietze extension theorem)**.** *Let X be normal and $A \subseteq X$ closed. Every continuous $f\colon A \to \mathbb{R}$ extends to a continuous $\tilde{f}\colon X \to \mathbb{R}$ with $\tilde{f}|_A = f$. If f takes values in $[-1, 1]$, then $\tilde{f}$ can be chosen with values in $[-1, 1]$.*

Proof. Step 1 (one approximation). Let $g\colon A \to [-M, M]$ be continuous. The sets $A_- = g^{-1}([-M, -M/3])$ and $A_+ = g^{-1}([M/3, M])$ are disjoint and closed in A, hence closed in X. Urysohn's lemma (rescaled) gives a continuous $h\colon X \to [-M/3, M/3]$ with $h = -M/3$ on A_- and $h = M/3$ on A_+. Checking the three regions of A separately shows

$$|g(a) - h(a)| \le \tfrac{2}{3}M \qquad \text{for all } a \in A.$$

Step 2 (bounded case). Let $f\colon A \to [-1, 1]$. Apply Step 1 to $g_0 = f$, $M = 1$ to get h_1 with $|h_1| \le \frac{1}{3}$ and $|f - h_1| \le \frac{2}{3}$ on A. Apply it to $g_1 = f - h_1$, $M = \frac{2}{3}$ to get h_2 with $|h_2| \le \frac{1}{3} \cdot \frac{2}{3}$ and

$|f - h_1 - h_2| \le (\tfrac{2}{3})^2$ on A. Inductively we obtain continuous $h_n \colon X \to \mathbb{R}$ with

$$|h_n| \le \tfrac{1}{3}\left(\tfrac{2}{3}\right)^{n-1} \text{ on } X, \qquad \Big|f - \sum_{k=1}^{n} h_k\Big| \le \left(\tfrac{2}{3}\right)^{n} \text{ on } A.$$

By the Weierstrass M-test, $\tilde{f} = \sum_{k\ge 1} h_k$ converges uniformly on X, so $\tilde{f}$ is continuous (Theorem 10.5). On A it equals f, and $|\tilde{f}| \le \tfrac{1}{3}\sum_{n\ge 1}(\tfrac{2}{3})^{n-1} = 1$.

Step 3 (unbounded case). Using a homeomorphism $\mathbb{R} \cong (-1,1)$ we may assume $f \colon A \to (-1,1)$. Step 2 gives a continuous $F \colon X \to [-1,1]$ extending f. The set $D = F^{-1}(\{-1,1\})$ is closed and disjoint from A. By Urysohn's lemma choose $\varphi \colon X \to [0,1]$ with $\varphi = 1$ on A and $\varphi = 0$ on D. Then $\tilde{f} = \varphi \cdot F$ is continuous, equals f on A, and takes values in $(-1,1)$: on D it is 0, and off D we have $|F| < 1$. $\square$

Chapter 17

Embeddings into Cubes and Metrization

Reading: Munkres, *Topology* (2nd ed.), §34.

Definition 17.1. Let $B \subseteq X$ be closed and $a \notin B$. A continuous $f\colon X \to \mathbb{R}$ *separates a from B* if $f(a) \notin \operatorname{cl}(f(B))$. A family $\mathcal{F}$ of continuous functions *separates points from closed sets* if for every closed B and $a \notin B$ some $f \in \mathcal{F}$ separates a from B.

Remark 17.2. If X is completely regular, then $C(X,[0,1])$ separates points from closed sets: choose f with $f(a) = 0$ and $f(B) \subseteq \{1\}$.

Theorem 17.3 (Embedding theorem)**.** *Let X be a space in which singletons are closed, and let $\mathcal{F} = \{f_j : j \in J\} \subseteq C(X,[0,1])$ separate points from closed sets. Then*

$$\psi\colon X \to [0,1]^J, \qquad \psi(x) = \big(f_j(x)\big)_{j\in J},$$

is an embedding.

Proof. We check three things.

(1) *ψ is injective.* Let $x \neq y$. Since $\{y\}$ is closed, some f_j separates x from $\{y\}$: $f_j(x) \notin \operatorname{cl}\{f_j(y)\} = \{f_j(y)\}$. So $\psi(x) \neq \psi(y)$.

(2) *ψ is continuous,* since each coordinate $\pi_j \circ \psi = f_j$ is continuous.

(3) *$\psi\colon X \to \psi(X)$ is open.* Let $U \subseteq X$ be open and $x \in U$. Choose j such that f_j separates x from the closed set $X \setminus U$, and let $I \ni f_j(x)$ be an open interval with $I \cap \operatorname{cl}\big(f_j(X \setminus U)\big) = \varnothing$. Let $W = \pi_j^{-1}(I)$, which is open in $[0,1]^J$. Then $\psi(x) \in W \cap \psi(X)$, and

$$W \cap \psi(X) \subseteq \psi(U)\,:$$

if $z = \psi(z') \in W$, then $f_j(z') \in I$, so $z' \notin X \setminus U$, i.e. $z' \in U$. Hence $\psi(U)$ is a union of sets open in $\psi(X)$.

Together these say $\psi\colon X \to \psi(X)$ is a homeomorphism. □

Proposition 17.4. *If X is regular and second countable, there is a* countable *family $\mathcal{F} \subseteq C(X,[0,1])$ that separates points from closed sets.*

Proof. By Theorem 15.7(6), X is normal. Let $\mathcal{B}$ be a countable basis and

$$J = \{(A,B) \in \mathcal{B}\times\mathcal{B} :\ \operatorname{cl}(A) \subseteq B\},$$

a countable set. For $(A,B) \in J$, Urysohn's lemma applied to the disjoint closed sets $\operatorname{cl}(A)$ and $X \setminus B$ gives $f_{A,B} \in C(X,[0,1])$ with $f_{A,B} = 0$ on $\operatorname{cl}(A)$ and $f_{A,B} = 1$ off B.

Claim. $\mathcal{F} = \{f_{A,B} : (A, B) \in J\}$ separates points from closed sets.

Let Q be closed and $p \notin Q$. Choose $B \in \mathcal{B}$ with $p \in B \subseteq X \setminus Q$. By regularity (Theorem 16.2 applied to $\{p\}$, or directly) there is an open U with $p \in U \subseteq \mathrm{cl}(U) \subseteq B$, and then $A \in \mathcal{B}$ with $p \in A \subseteq U$. So $\mathrm{cl}(A) \subseteq B$, i.e. $(A, B) \in J$. Now $f_{A,B}(p) = 0$ while $f_{A,B}(Q) \subseteq \{1\}$, so $f_{A,B}(p) \notin \mathrm{cl}(f_{A,B}(Q))$. □

Corollary 17.5 (Urysohn metrization theorem)**.** *Every regular second countable space is metrizable.*

Proof. By Theorems 17.3 and 17.4, X embeds into $[0, 1]^{\mathbb{N}}$, which is metrizable (Theorem 9.9). A subspace of a metrizable space is metrizable. □

Chapter 18

The Baire Category Theorem

Reading: Munkres, *Topology* (2nd ed.), §§48, 49.

Baire's theorem says that in a complete metric space, or in a compact Hausdorff space, "countably many small sets cannot fill the space". Its strength lies in existence proofs: to show some object with a property exists, show that the objects *without* it form a countable union of thin sets.

18.1 Baire spaces

Definition 18.1. A subset A of a space X is *nowhere dense* if $\operatorname{int}(\operatorname{cl}(A)) = \varnothing$. A countable union of nowhere dense sets is called *meagre* (or of *first category*).

Example 18.2.

- In $\mathbb{R}$: any finite set, $\mathbb{Z}$, $\{1/n : n \in \mathbb{N}\}$ and the Cantor set are nowhere dense. $\mathbb{Q}$ is *not* nowhere dense ($\operatorname{cl}\mathbb{Q} = \mathbb{R}$), but it is meagre, being a countable union of singletons.
- A closed set A is nowhere dense iff $\operatorname{int}(A) = \varnothing$, iff its complement $X \setminus A$ is open and dense.

Definition 18.3 (Baire space; Munkres §48)**.** A space X is a *Baire space* if the following holds: whenever $\{A_n\}_{n\in\mathbb{N}}$ is a countable family of closed sets each with empty interior, their union $\bigcup_n A_n$ also has empty interior.

Example 18.4.

(1) $\mathbb{Q}$ is *not* a Baire space: each $\{q\}$ is closed with empty interior in $\mathbb{Q}$, but $\bigcup_{q\in\mathbb{Q}}\{q\} = \mathbb{Q}$ has non-empty interior.

(2) Any discrete space, e.g. $\mathbb{N}$, *is* a Baire space: every non-empty set has non-empty interior, so the only closed set with empty interior is $\varnothing$.

(3) The subspace $\{0\} \cup \{1/n : n \in \mathbb{N}\}$ of $\mathbb{R}$ is a Baire space: every non-empty open set contains some isolated point $1/n$, so a set with empty interior misses all $1/n$, i.e. is $\varnothing$ or $\{0\}$.

Lemma 18.5 (Munkres, Lemma 48.1)**.** *X is a Baire space if and only if for every countable family $\{U_n\}$ of open dense subsets of X, the intersection $\bigcap_n U_n$ is dense.*

Proof. Use complements. U is open and dense $\iff$ $A = X \setminus U$ is closed with empty interior (since $X \setminus \operatorname{cl}(U) = \operatorname{int}(X \setminus U)$). Also $\bigcap_n U_n$ is dense $\iff$ $X \setminus \bigcap_n U_n = \bigcup_n A_n$ has empty interior. So the two conditions say the same thing. □

Remark 18.6. A countable intersection of open sets is called a G_δ *set*. Theorem 18.5 says: in a Baire space, a countable intersection of dense open sets is a dense G_δ. Such sets are "large"; their complements are meagre.

18.2 The Baire category theorem

Lemma 18.7 (Nested closed sets)**.** *Let (X, d) be complete and $C_1 \supseteq C_2 \supseteq \cdots$ non-empty closed sets with $\operatorname{diam}(C_n) \to 0$. Then $\bigcap_n C_n$ is a single point.*

Proof. Pick $x_n \in C_n$. For $m, n \geq N$ both $x_m, x_n \in C_N$, so $d(x_m, x_n) \leq \operatorname{diam}(C_N) \to 0$: the sequence is Cauchy and converges to some x. For each N, the tail $(x_n)_{n \geq N}$ lies in the closed set C_N, so $x \in C_N$. If y is also in every C_N, then $d(x, y) \leq \operatorname{diam}(C_N) \to 0$, so $y = x$. □

Theorem 18.8 (Baire category theorem; Munkres, Theorem 48.2)**.** *If X is a compact Hausdorff space or a complete metric space, then X is a Baire space.*

Proof. Let $A_1, A_2, \ldots$ be closed sets with empty interior, and let $U_0 \neq \varnothing$ be open. We must find a point of U_0 outside every A_n.

Key step. Given a non-empty open U and a closed A with empty interior, $U \not\subseteq A$, so $U \setminus A$ is open and non-empty. Since X is regular (compact Hausdorff spaces are normal, metric spaces are normal), choose a non-empty open V with

$$\operatorname{cl}(V) \subseteq U \setminus A.$$

In the metric case, also take V to be a small ball, so that $\operatorname{diam}(V) < 1/n$ at stage n.

Apply the key step repeatedly: get non-empty open sets $U_1, U_2, \ldots$ with

$$\operatorname{cl}(U_n) \subseteq U_{n-1} \setminus A_n \qquad (n \geq 1),$$

and in the metric case $\operatorname{diam}(U_n) < 1/n$. The closed sets $C_n = \operatorname{cl}(U_n)$ are non-empty and nested.

- *Compact Hausdorff case:* $\{C_n\}$ has the finite intersection property, so $\bigcap_n C_n \neq \varnothing$ (Theorem 14.27).
- *Complete metric case:* $\operatorname{diam}(C_n) \leq 1/n \to 0$, so $\bigcap_n C_n \neq \varnothing$ by Theorem 18.7.

Any $x \in \bigcap_n C_n$ lies in $C_1 \subseteq U_0$ and, for every n, in $C_n \subseteq X \setminus A_n$. So $U_0 \not\subseteq \bigcup_n A_n$, i.e. $\bigcup_n A_n$ has empty interior. □

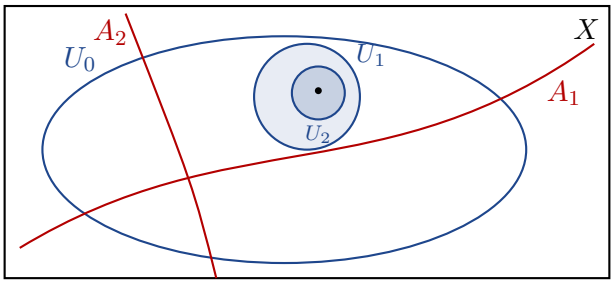


Figure 18.1: Shrinking open sets that dodge $A_1, A_2, \ldots$ one at a time; their closures meet in a point avoiding all A_n.

Lemma 18.9 (Munkres, Lemma 48.4)**.** *Every open subspace Y of a Baire space X is a Baire space.*

Proof. Let A_n be closed in Y with empty interior in Y. Then $\mathrm{cl}_X(A_n) \cap Y = A_n$.

Claim: $\mathrm{cl}_X(A_n)$ *has empty interior in* X. If W is open in X with $W \subseteq \mathrm{cl}_X(A_n)$, then $W \cap Y$ is open in Y and $W \cap Y \subseteq \mathrm{cl}_X(A_n) \cap Y = A_n$, so $W \cap Y = \varnothing$. But $W \subseteq \mathrm{cl}_X(A_n) \subseteq \mathrm{cl}_X(Y)$, and a non-empty open set inside $\mathrm{cl}_X(Y)$ must meet Y. Hence $W = \varnothing$.

Since X is Baire, $\bigcup_n \mathrm{cl}_X(A_n)$ has empty interior in X. If $V \subseteq \bigcup_n A_n$ were open in Y, it would also be open in X (as Y is open), and contained in $\bigcup_n \mathrm{cl}_X(A_n)$; so $V = \varnothing$. □

Corollary 18.10. *Every locally compact Hausdorff space is a Baire space.*

Proof. If X is compact this is Theorem 18.8. Otherwise X is an open subspace of its one-point compactification X^* (it is the complement of the closed point ∞), which is compact Hausdorff, hence Baire. Now apply Theorem 18.9. □

18.3 First applications

Corollary 18.11. *A non-empty complete metric space (or compact Hausdorff space) with no isolated points is uncountable. In particular* $\mathbb{R}$, $[0,1]$ *and the Cantor set are uncountable.*

Proof. If $X = \{x_1, x_2, \dots\}$ were countable, each $\{x_n\}$ would be closed (Hausdorff) with empty interior (not isolated), yet their union X has non-empty interior. □

Example 18.12 ($\mathbb{Q}$ is not a G_δ in $\mathbb{R}$)**.** Suppose $\mathbb{Q} = \bigcap_n U_n$ with U_n open in $\mathbb{R}$. Each $U_n \supseteq \mathbb{Q}$ is dense. For each $q \in \mathbb{Q}$, $\mathbb{R} \setminus \{q\}$ is open and dense. Then

$$\bigcap_n U_n \;\cap\; \bigcap_{q \in \mathbb{Q}} (\mathbb{R} \setminus \{q\}) = \mathbb{Q} \setminus \mathbb{Q} = \varnothing$$

is a countable intersection of dense open sets, which must be dense by Theorems 18.5 and 18.8. Contradiction. Consequently there is no function $f\colon \mathbb{R} \to \mathbb{R}$ continuous exactly at the rationals (the continuity set of any function is always a G_δ).

Example 18.13 (The irrationals are a Baire space)**.** $\mathbb{R} \setminus \mathbb{Q} = \bigcap_{q \in \mathbb{Q}} (\mathbb{R} \setminus \{q\})$ is a dense G_δ in $\mathbb{R}$. A dense G_δ subspace of a Baire space is Baire (an exercise in Munkres §48). Alternatively, $\mathbb{R} \setminus \mathbb{Q} \cong \mathbb{N}^{\mathbb{N}}$, which is completely metrizable.

Remark 18.14. Being Baire is a topological property, but completeness is not: $(0,1) \cong \mathbb{R}$ is not complete in the usual metric, yet it is Baire. The relevant notion is *completely metrizable*: the topology comes from *some* complete metric.

18.4 Limits of continuous functions

Theorem 18.15 (Munkres, Theorem 48.5)**.** *Let X be a Baire space and Y a metric space. Let $f_n\colon X \to Y$ be continuous and suppose $f_n(x) \to f(x)$ for every $x \in X$. Then the set of points at which f is continuous is dense in X.*

Proof. For $N \in \mathbb{N}$ and $\varepsilon > 0$ let

$$A_N(\varepsilon) = \{x \in X : \; d(f_n(x), f_m(x)) \le \varepsilon \text{ for all } n, m \ge N\}.$$

This is closed, being an intersection of the closed sets $\{x : d(f_n(x), f_m(x)) \le \varepsilon\}$ (preimages of $[0,\varepsilon]$ under continuous maps). Since $(f_n(x))$ converges, it is Cauchy, so $X = \bigcup_N A_N(\varepsilon)$.

Claim 1. $U(\varepsilon) = \bigcup_N \operatorname{int}(A_N(\varepsilon))$ is open and dense in X.

Proof. Openness is clear. Let $V \neq \varnothing$ be open. By Theorem 18.9, V is Baire, and $V = \bigcup_N (A_N(\varepsilon) \cap V)$ is a countable union of closed subsets of V, so some $A_N(\varepsilon) \cap V$ has non-empty interior in V. That interior is open in X and lies in $\operatorname{int}(A_N(\varepsilon))$, so V meets $U(\varepsilon)$.

Claim 2. If $x_0 \in U(\varepsilon)$, there is an open $W \ni x_0$ with $d(f(x), f(x_0)) \le 3\varepsilon$ for all $x \in W$.

Proof. Say $x_0 \in \operatorname{int}(A_N(\varepsilon))$. Letting $m \to \infty$ in the definition of $A_N(\varepsilon)$ gives $d(f_N(x), f(x)) \le \varepsilon$ for all $x \in A_N(\varepsilon)$. By continuity of f_N, choose an open $W \ni x_0$ with $W \subseteq \operatorname{int}(A_N(\varepsilon))$ and $d(f_N(x), f_N(x_0)) < \varepsilon$ on W. Then for $x \in W$,

$$d(f(x), f(x_0)) \le d(f(x), f_N(x)) + d(f_N(x), f_N(x_0)) + d(f_N(x_0), f(x_0)) \le 3\varepsilon.$$

Now let $C = \bigcap_{k\in\mathbb{N}} U(1/k)$. By Theorem 18.5 C is dense, and by the second claim f is continuous at every point of C. □

Corollary 18.16. *If $g\colon \mathbb{R} \to \mathbb{R}$ is differentiable, then g' is continuous on a dense set.*

Proof. $g'(x) = \lim_n n(g(x + \frac{1}{n}) - g(x))$ is a pointwise limit of continuous functions, and $\mathbb{R}$ is Baire. □

18.5 A continuous nowhere-differentiable function

Let $I = [0,1]$ and let $\mathcal{C} = C(I, \mathbb{R})$ with the sup metric $\rho(f,g) = \sup_{x\in I} |f(x) - g(x)|$. This metric space is *complete*: a Cauchy sequence converges uniformly, and the limit is continuous by Theorem 10.5. So $\mathcal{C}$ is a Baire space.

Theorem 18.17 (Munkres, Theorem 49.1)**.** *Let $h \in \mathcal{C}$ and $\varepsilon > 0$. There is $g \in \mathcal{C}$ with $\rho(g,h) < \varepsilon$ that is differentiable at no point of I. In fact, the nowhere-differentiable functions contain a dense G_δ subset of $\mathcal{C}$.*

Proof. For $n \in \mathbb{N}$ let U_n be the set of $f \in \mathcal{C}$ such that

$$\forall x \in I \;\; \exists h \text{ with } 0 < |h| \le \tfrac{1}{n},\; x + h \in I : \quad |f(x+h) - f(x)| > n|h|.$$

In words: at every point, f has a steep secant (slope $> n$) of width at most $1/n$.

Claim 1. If $f \in \bigcap_n U_n$, then f is differentiable at no point of I.

Proof. Suppose f is differentiable at x (one-sided at an endpoint). Then the quotients $|f(x+h) - f(x)|/|h|$ stay bounded for $0 < |h| < \delta$, say by K, for some $\delta > 0$. For $|h| \ge \delta$ they are at most $2\rho(f,0)/\delta$. So all of them are bounded by some L. Choosing $n > L$, the condition $f \in U_n$ gives a quotient $> n > L$: contradiction.

Claim 2. U_n is open in $\mathcal{C}$.

Proof. Let $f \in U_n$. For each $x \in I$ the set of admissible h that work for x is relatively open and non-empty (by continuity of f), and a non-empty relatively open subset of an interval contains interior points, so we may pick h_x working for x with $x + h_x$ in the open interval $(0,1)$ and $|h_x| < 1/n$. Let $\varepsilon_x = |f(x+h_x) - f(x)| - n|h_x| > 0$. By continuity of f, the set

$$W_x = \{x' \in I :\; x' + h_x \in (0,1),\; |f(x'+h_x) - f(x')| - n|h_x| > \varepsilon_x/2\}$$

is open in I and contains x. By compactness, $I = W_{x_1} \cup \cdots \cup W_{x_k}$. Let $\delta = \frac{1}{4} \min_i \varepsilon_{x_i}$. If $\rho(g, f) < \delta$ and $x' \in W_{x_i}$, then with $h = h_{x_i}$,

$$|g(x' + h) - g(x')| \geq |f(x' + h) - f(x')| - 2\delta > n|h| + \tfrac{\varepsilon_{x_i}}{2} - 2\delta \geq n|h|.$$

So $g \in U_n$; the δ-ball around f lies in U_n.

Claim 3. U_n is dense in $\mathcal{C}$.

Proof. Let $f \in \mathcal{C}$ and $\varepsilon > 0$. Since f is uniformly continuous, it is within $\varepsilon/2$ of a piecewise linear g (join the points $(k/m, f(k/m))$ for large m). Let M be the largest absolute slope of g. Let s be a "sawtooth" function: piecewise linear, with values in $[0, \varepsilon/2]$ and all slopes equal to $\pm m$, where $m > M + n$. Then $g + s \in \mathcal{C}$ and $\rho(g + s, f) < \varepsilon$.

Given $x \in I$, both g and s are linear on some one-sided interval $[x, x + h]$ or $[x - h, x]$ inside I (there are finitely many corners), and we may take $0 < h \leq 1/n$. On that interval $g + s$ has slope of absolute value at least $m - M > n$, so this h works for x. Thus $g + s \in U_n$.

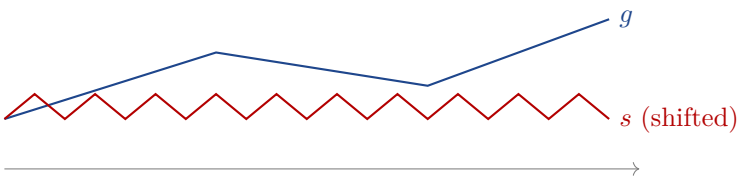


Figure 18.2: A piecewise-linear g plus a small but very steep sawtooth s has steep secants everywhere.

Since $\mathcal{C}$ is Baire, $\bigcap_n U_n$ is a dense G_δ (Theorem 18.5), so it contains some g with $\rho(g, h) < \varepsilon$. By Claim 1, g is nowhere differentiable. □

Remark 18.18. The proof shows that "most" continuous functions, in the sense of Baire category, are nowhere differentiable, even though the functions we usually write down are smooth almost everywhere. Explicit examples exist (Weierstrass: $\sum_n a^n \cos(b^n \pi x)$ with $0 < a < 1$, $ab > 1 + \frac{3\pi}{2}$, b an odd integer), but Baire's theorem gives existence with almost no computation.

References

- J. R. Munkres, *Topology*, 2nd ed., Prentice Hall, 2000. The main reference for these notes; each chapter lists the matching sections.